TEST REPORT
UL 62841-1
Electric Motor-Operated Hand-Held Tools, Transportable Tools and Lawn and Garden Machinery – Safety - Part1: General Requirements

Report Number.....:	XK2407012054S
Tested by (name + signature) .....:	Smile Xu
Compiled by (name + signature) ..:	Nina Li
Approved by (name + signature) ..	Andy Wang
Date of issue.....:	July 26, 2024
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Name of Testing Laboratory preparing the Report .....:	**Shenzhen SiCT Technology Co., Ltd.** 202, Building 3, No.111 Huanguan Middle Road, Songyuanxia Community, Guanhu Street, Longhua District, Shenzhen, Guangdong, China
Applicant's name .....:	**YongKang YeKaiAbrasive Co., Ltd**
Address.....:	Building 1, NO.158, ChangSheng East Road, Xiangzhu Village, XiangzhuTown, YongKang, Jinhua, Zhejiang, China
Manufacturer's name.....:	**YongKang YeKaiAbrasive Co., Ltd**
Address.....:	Building 1, NO.158, ChangSheng East Road, Xiangzhu Village, XiangzhuTown, YongKang, Jinhua, Zhejiang, China
Test specification:	
Standard.....:	UL 62841-1, Edition 1, Edition date: February 20, 2015, Re: August 21, 2020
Test procedure .....:	UL test report
Non-standard test method .....:	N/A
TRF template used.....:	UL62841-1
Test Report Form(s) Originator ....:	SiCT
Master TRF.....:	2020-12-03

Test item description.....:	Steel wire wheel
Trade Mark.....:	N/A
Model/Type reference.....:	8", 6", 10"
Ratings.....:	SIZE: 8"

List of Attachments (including a total number of pages in each attachment):

- Main report (88 pages): UL 62841-1 TRF
- Appendix 1 (2 pages): Product photos

Summary of testing:

The sample(s) tested complies with the requirements of UL 62841-1.

The model no. 8" was selected as representative for full test unless other specified, and complied with the requirements of standard. All models are identical except the model name and appearance.

Tests performed (name of test and test clause):	**Testing location:**
According to the General product information, Full tests were performed on model 8".	**Shenzhen SiCT Technology Co., Ltd.** 202, Building 3, No.111 Huanguan Middle Road, Songyuanxia Community, Guanhu Street, Longhua District, Shenzhen, Guangdong, China

Copy of marking plate:

The artwork below may be only a draft.

Steel wire wheel

Model: 8”

SIZE: 8"



Manufacturer: YongKang YeKaiAbrasive Co., Ltd

Made in China

Marking label location

Notes:

- The above markings are the minimum requirements required by the safety standard as a reference marking label. For the final production samples, the additional markings which do not give rise to misunderstanding may be added.
- The model’s name can be replaced by other’s model in this report.

Test item particulars..... :	Steel wire wheel
Category of equipment ..... :	~~Hand held~~, ~~Transportable,~~ Lawn / Garden
Protection Class of tool ..... :	~~Class I, Class II, Class III~~
Method of supply cord attachment ..... :	~~Type X, Type Y, Type Z, appliance inlet~~
Duty conditions..... :	Normal,~~ severe, extra severe~~
Type of operation..... :	Normal~~, short-time, intermittent~~
Degree of protection..... :	N/A
Accessories and detachable parts included..... :	N/A
Other options included ..... :	
Classification of installation and use..... :	N/A
Supply Connection ..... :	N/A
Possible test case verdicts:	
- test case does not apply to the test object.....:	N/A
- test object does meet the requirement.....:	P (Pass)
- test object does not meet the requirement.....:	F (Fail)
Testing.....:	
Date of receipt of test item .....:	July 19, 2024
Date (s) of performance of tests .....:	July 20, 2024 – July 26, 2024

General remarks:

"(See Enclosure #)" refers to additional information appended to the report.
"(See appended table)" refers to a table appended to the report.

Throughout this report a ☐ comma / ☒ point is used as the decimal separator.

General product information and other remarks:

1. The appliance is a Steel wire wheel intend for household use.
2. The appliance is a non-electrical part on the lawn mower.

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Clause	Requirement + Test	Result - Remark	Verdict

5	**GENERAL CONDITIONS FOR THE TESTS**		P
5.1	General test conditions in this clause apply unless otherwise specified in this standard		P
5.2	Tests made on separate samples		P
	At manufacturer's discretion, fewer samples used		P
	Cumulative stress from successive tests on electronic circuits avoided		P
	Several tests conducted on a single sample, results not affected by previous tests.		P
5.3	Evident from construction of the tool that a particular test(s) not applicable, test(s) not made..... :		P
5.4	Tests carried out with the tool and/or any movable part of it		P
	Tool placed in the most unfavourable position that may occur in normal use.		P
5.5	Tools provided with controls or switching devices and setting can be altered by the user, controls or devices adjusted to their most unfavourable settings		N/A
	Electronic speed control devices set at their highest speed ..... :		N/A
	Adjusting means accessible without the aid of a tool, this subclause applies whether the setting can be altered by hand or with the aid of a tool. Adjusting means not accessible without the aid of a tool and setting is not intended to be altered by the user, this subclause does not apply.		N/A
	Adequate sealing prevents alteration of setting by user		N/A
5.6	Tests conducted in a draught-free location, and unless otherwise specified, in (20 ± 5) °C		P
	Tests conducted at (23 ± 2) °C due to temperature limited temperature sensitive device		N/A
5.7.1	Tools for a.c. only, tested with a.c. at rated frequency, if marked		N/A
	Tools marked for a.c./d.c., tested with the most unfavourable supply ..... :		P
	Tools for a.c. not marked with rated frequency, or marked 50-60 Hz or 50/60 Hz, tested with either 50 Hz or 60 Hz, whichever is the most unfavourable		N/A
	Tools with series motors only, either frequency may be used		N/A
5.7.2	Tool rated for more than one rated voltage or a voltage range, tested at the highest voltage (V)..... :		N/A

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Clause	Requirement + Test	Result - Remark	Verdict
5.7.3	Tools where there is no marked rated current , tests that require a value for rated current conducted at current measured rated input at the lowest rated voltage or the lower value of the rated voltage range		N/A
5.8	Alternative heating elements or attachments which are made available for the tool by manufacturer, tool is tested with those heating elements or attachments which give the most unfavourable results		N/A
5.9	Tools are tested with the specified flexible supply cord connected to the tool.		N/A
5.10	Parts of class I tool having accessible parts not connected to an earthing terminal or earthing contact, and not separated from live parts by an intermediate metal part connected to an earthing terminal/contact, were checked on class II construction requirements.		N/A
5.11	Class I tool or class II tool having parts operating at safety extra-low voltage, such parts on requirements specified for class III tools		N/A
5.12	When testing electronic circuits, supply is free from perturbations from external sources that can influence the results of the tests		N/A
5.13	Heating element, if any, cannot be operated unless the motor is running, element is tested with the motor running		N/A
	Heating element, if any, can be operated without the motor running, element is tested with or without the motor running, whichever is the more unfavourable		N/A
	Heating elements incorporated in the tool connected to a separate supply unless otherwise specified		N/A
5.14	For attachments performing a function within the scope of IEC 62841-2, IEC 62841-3 or IEC 62841-4, tests made in accordance with IEC 62841-2, IEC 62841-3 or IEC 62841-4.		P
5.15	Method of torque loading chosen so as to avoid additional stresses, such as by side thrust.		P
	Additional loads necessary for the correct operation of the tool considered ..... :		P
	Brake used for loading, load applied gradually		P
	Modification of output means for purpose of loading permitted to allow connection to brake		P
5.16	Tools intended for SELV tested using a supply transformer intended to be used with the tool.		N/A
5.17	For requirements based on the mass of the tool, the mass is determined without supply cord and without tool bits or accessories, but with all equipment and attachments needed for normal use		P

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Clause	Requirement + Test	Result - Remark	Verdict
	Required accessories, equipment and attachments as given in the relevant part of IEC 62841-2, IEC 62841-3 or IEC 62841-4.		P
	If tool has more accessories, equipment or attachments heaviest configuration shall be used to determine mass.		P
5.18	For linear and angular dimensions, ISO 2768-1, class “c” applicable, unless tolerances are specified		P
5.19	All electrical measurements made with a maximum measurement error of 5 %.		P
	Instruments for measuring voltage have input resistance ≥ 1 MΩ and parallel capacitance ≤ 150 pF.		N/A
5.20	Thermal equilibrium considered achieved when the total deviation of three successive temperature readings, taken at 3 min intervals, is ≤ 4 K		N/A
	Induction motor, measurement time of 1 hour is considered sufficient.		N/A
6	**RADIATION, TOXICITY AND SIMILAR HAZARDS**		P
6.1	No harmful radiation, no toxic or similar hazard		P
6.2	For tool with laser to indicate a cutting line or the like, laser class 2M or lower according to IEC 60825-1:2007.	No laser used	N/A
	Tool marked with symbol(s) as in of IEC 60825-1: 2007 for the relevant laser class.		N/A
6.3	Tool fitted with non-coherent light sources, users of tools are cautioned as to the risk of potential photo-biological harm, if such harm exist..... :		N/A
6.3.1	Visible light indicators (pilot lamps) and Infrared sources used for signalling and communication considered to have no risk of photo-biological harm, no marking required.		N/A
6.3.2	Tools emitting visible light from electroluminescent, incandescent or LED sources, considered to be for short term, non-general light services use where exposure is both incidental and intermittent	No LED used	N/A
	Marked with either: – “CAUTION Do not stare at operating lamp”, or – symbol 60417-6041(2010-08)		N/A
	If no reasonable risk of harm, markings may be omitted		N/A

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Clause	Requirement + Test	Result - Remark	Verdict
	No reasonable risk of harm considered, as either a) light emission at a distance of 200 mm along any direction of the tool < 500 Lux; or b) luminance light emission < 10 000 cd/m² in the range of visible light; or c) light source (if not focused by external optics) is in Risk Group 1 or lower evaluated by the methods of IEC 62471; or d) tool itself evaluated by the methods of IEC 62471 and found to be in Risk Group 1 or lower.		N/A
6.3.3	For light derived by sources other than those mentioned in 6.3.2, product evaluated by the methods of IEC 62471, markings guided by 5.4 of IEC/TR 62471-2:2009.		N/A
7	**CLASSIFICATION**		N/A
7.1	Tool is Class I, II, or III with respect to protection against electric shock.....:		N/A
7.2	Degree of protection against harmful ingress of water per IEC 60529 .....:		N/A
	Required degree of protection other than IPX0 specified in relevant part of IEC 62841-2, IEC 62841-3 or IEC 62841-4.....:		N/A
8	**MARKINGS AND INSTRUCTIONS**		P
8.1	Tool marked with rated voltage(s) or rated voltage range(s) (V)..... :	See the marking plate	N/A
	Tool for star-delta connection clearly marked with the two rated voltages (e.g. 230 Δ / 400 Y V)..... :		N/A
	Tool complying with this standard for a voltage range, may be marked with any single voltage or smaller voltage range within that range (V)..... :		N/A
	Symbol for nature of supply or rated frequency or frequency range. The symbol for nature of supply placed next to rated voltage (Hz) ..... :		N/A
	Rated input or current marked (W or A) ..... :	See the marking plate	N/A
	Tool has alternative components to be selected by a control device, rated input or rated current is that corresponding to the highest rated input or rated current		N/A
	Class II symbol for class II tools..... :		N/A
	IP number other than IPX0..... :		N/A
8.1.1	Tools with range of rated values (e.g. voltage, frequency) can be operated without adjustment over the range, marked with the lower and upper limits of the range separated by a hyphen, e.g. 115-230 V. :		N/A

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Clause	Requirement + Test	Result - Remark	Verdict
	Different rated values to be adjusted by the user / installer, tool marked with the these values separated by an oblique stroke, e.g. 115/230 V..... :		N/A
8.1.2	Upper and lower limits of rated power input marked,		N/A
	unless difference between upper and lower limits of rated voltage range do not exceed 20 % of the mean value, in which case the rated input is related to mean value of voltage range.		N/A
8.2	Tool marked with - "WARNING – To reduce the risk of injury, user must read instruction manual", or - sign M002 of ISO 7010, or - appropriate symbol, see relevant part of IEC 62841-2, IEC 62841-3 or IEC 62841-4		P
	"WARNING" in capital letters not less than 2,4 mm high, not separated from either the cautionary statement or the symbol ISO 7000-0434A or ISO 7000-0434B		N/A
	Statement verbatim except that "operator's manual" or "user guide" may replace "instruction manual".		P
	Additional symbols in accordance with ISO 7010 or designed in accordance with ISO 3864-2/3864-3. . :		P
	Cautionary statements having the same signal word such as "WARNING" may be combined into one paragraph under one signal word		N/A
	Order of statements: markings required by Part 1, markings required by part of IEC 62841-2, IEC 62841-3 or IEC 62841-4 and then any optional markings		N/A
8.3	Business name and address of manufacturer, at least country or state, city and postal code ..... :	See the marking plate	P
	Business name and address of authorized representative, at least country or state, city and postal code ..... :	See the marking plate	P
	Designation of the tool (may be coded)..... :		N/A
	Designation coded, code explained in the instructions		N/A
	Designation of series or type..... :		N/A
	Year of manufacture and a date code identifying at least the month of manufacture ..... :		P
	Tools parts shipped separately for assembly by the end user, each part marked for identification on the part or the package		N/A
	"> 25 kg" if the mass of the tool is over 25 kg		N/A
	No misunderstanding through additional markings		N/A
8.4	Markings of 8.1 to 8.3 not on detachable part of the tool		P

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Clause	Requirement + Test	Result - Remark	Verdict
	Markings of 8.2 and 8.3 clearly discernible from outside the tool		P
	Markings other than symbols, fold-over label on power cords used (Y or Z attachments only)		N/A
	Other markings may be visible after removing cover		N/A
	Indications for switches and controls placed on or in vicinity of components		P
	Not placed on parts which can be repositioned		P
	Not positioned such that the marking is misleading		P
8.5	Tool can be adjusted to suit different rated voltages, change in voltage clearly discernible		N/A
	Correct Wiring diagram fixed to tool, may be on inside of a cover but not on a label loosely attached to the tool		N/A
8.6	Use of correct units		P
	Use of correct symbols		P
	Additional symbols explained in the instructions, no misunderstanding		P
	Other units and their symbols belong to the international standardized system.		P
	Other units and their symbols same as international standardised system ..... :		P
8.7	Connection diagram affixed to tool with more than two supply conductors, unless terminals clearly identified		N/A
	The earthing conductor not a supply conductor		N/A
	Wiring diagram indicates how the windings are to be connected for tools for star-delta connection		N/A
8.8	Terminals, except for type Z attachments, marked on non-removable part with specified symbols:		-
	- Terminal exclusively for neutral connection marked with “N”		N/A
	- Earthing terminal marked with symbol IEC 60417-5019 (2006-08)		N/A
	The markings not placed on screws, removable washers or other parts which might be removed		N/A
8.9	Switches which may result in a hazard marked or placed to indicate which part of tool they control.... :		P
8.10	“Off” position of multi stable power switch indicated by figure O (symbol of IEC 60417-5008 [2002-10])		P
	A momentary power switch which can be locked in the “on” position is not considered as a multi -stable switch.		P
	Push-buttons for "off" function only, figure O used, button coloured red or black..... :		P

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Clause	Requirement + Test	Result - Remark	Verdict
	Figure O not used for any other indication		P
	Transportable tools, power switch actuator or cover not coloured yellow and red as specified for emergency stop according to ISO 13850.		N/A
	Flap/cover covers only the start button, colour of the flap/cover not black, red or yellow..... :		N/A
	Flap/cover covers only the stop button, colour of the flap/cover red or yellow ..... :		N/A
8.11	Control devices adjusted during operation and the like provided with markings as specified, unless…		N/A
	… fully “on” position opposite to “off” position		N/A
	Figures used for different positions with O for “off” position, and figures reflecting greater output for other positions ..... :		N/A
	Indication for different positions placed on the device itself, or adjacent to the operating means		N/A
8.12	Markings easily legible		P
	Markings withstood durability test: - 15 s with water soaked cloth - 15 s with petroleum spirit soaked cloth		P
	Signs are in contrast to their background, clearly legible from a distance of not less than 500 mm		P
	Effect of normal use taken into account		P
	Adhesive backing durable, meets requirements of UL 969 or…		N/A
	… withstands specified tests	See tables 8.12 A - D	P
8.13	Thermal link or fuse-link, reference number or other means for identifying the link marked		N/A
8.14	Instruction manual and safety instructions: - are provided together with the tool		P
	- are noticed by the user when the tool is removed from the packaging		P
	- include an explanation of the symbols		P
	- are written in the official language(s) of the country in which the tool is sold.....:		P
	- are legible and contrast with the background.		P
	- include business name and address of the manufacturer and, where applicable, his authorised representative .....:		P
	- include the designation of the tool and series or type as required by 8.3, including description of machine such as “drill”, “planer” etc. .....:		P
8.14.1	Safety instructions in English are verbatim and in any other official language are equivalent..... :		P

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Clause	Requirement + Test	Result - Remark	Verdict
	The general power tool safety warnings may be separate from the instruction manual.		P
	Term “tool” or “power tool” not used for garden machinery; use term such as “machine”		P
	Format of all Safety Warnings differentiate the context of all clauses by font or similar means and as illustrated in 8.14.1.1		P
8.14.1.1	General Power Tool Safety Warnings		P
	1) Work Area Safety		P
	2) Electrical Safety		P
	3) Personal Safety		P
	4) Power Tool Use and Care		P
	5) Service		P
8.14.1.2	Order of the Safety Instructions in accordance with A): Part 1 warnings are followed by the relevant part of IEC 62841-2, IEC 62841-3 or IEC 62841-4 warnings, or …		P
	B): Part 1 and part 2, 3 or 4 warnings divided into the sections defined by the numbered subtitles and the associated warnings below the numbered subtitle		P
	Format of instruction manual section titles for IEC 62841-2, IEC 62841-3 or IEC 62841-4 warnings		P
	C): Any additional warnings deemed necessary by the manufacturer, not inserted within any of the IEC 62841-1, IEC 62841-2, IEC 62841-3 or IEC 62841-4 warnings		P
8.14.1.3	Instruction manual and safety instructions in one common document, or		P
	Warning as specified included in manual		P
8.14.2	Additional instructions and information		P
	a) Instructions for putting into use		P
	b) Operating instructions		P
	c) Maintenance and servicing instructions		P
	d) Warnings and instructions for tools with a liquid system		P
8.14.3	Information about the mass or weight of the tool, if any, is the mass specified in 5.17.		P
9	**PROTECTION AGAINST ACCESS TO LIVE PARTS**		N/A
9.1	Tools so constructed and enclosed that there is adequate protection against accidental contact with live parts, even after removal of detachable parts and soft materials		N/A
9.2	Accessible part not considered live if it is:		N/A

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Clause	Requirement + Test	Result - Remark	Verdict
	- supplied with SELV		N/A
	- or separated from live parts by protective impedance, d.c. current not exceeding 2 mA		N/A
	- or separated from live parts by protective impedance, a.c. peak value not exceeding 0.7 mA		N/A
	- for peak value 42.4 V up to and including 450 V capacitance not exceeding 0.1 μF		N/A
	- for peak value 450 V up to and including 15 kV discharge not exceeding 45 μF		N/A
9.3	Lamps located behind a detachable cover are not removed		N/A
	Protection against contact with live parts of the lamp cap ensured during insertion or removal of lamps located behind a detachable cover		N/A
	Test probe B of IEC 61032:1997 applied with a force of ≤5 N		N/A
	Opening does not allow entry of test probe B of IEC 61032:1997, rigid test probe applied with a force of 20 N		N/A
	Test with probe B of IEC 61032:1997 repeated		N/A
	Test probe does not touch live parts or live parts protected only by lacquer, enamel, ordinary paper, cotton, oxide film, beads or sealing compound		N/A
9.4	Test probe 13 of IEC 61032:1997 applied with a force ≤5 N through openings in class II tools and class II constructions		N/A
	Exception: openings giving access to lamp caps and live parts in socket-outlets		N/A
	Test probe is also applied through openings in earthed metal enclosures having a non-conductive coating such as enamel or lacquer.		N/A
	Not be possible to touch live parts with the test probe		N/A
9.5	Class II tools and class II constructions, adequate protection against accidental contact with basic insulation and metal parts separated from live parts by basic insulation only		N/A
	Parts not separated from live parts by double or reinforced insulation are not accessible		N/A
	Probe B of IEC 61032:1997 cannot contact basic insulation through openings in Class II tools or Class II constructions		N/A
10	**STARTING**		N/A
10.1	Motors start under normal voltage conditions		N/A
	Starting ten times at 0.85 times rated voltage without load (V) ..... :		N/A

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Clause	Requirement + Test	Result - Remark	Verdict
	Starting ten times at 1.1 times rated voltage without load (V) ..... :		N/A
	Tool operated and overload protection devices incorporated in the tool did not activate.		N/A
	Centrifugal and other automatic starting switches operate reliably and without contact chattering		N/A
10.2	Input current drawn at (2,0 ±0,2) s after starting does not exceed 30 A…		N/A
	… or 4 times the rated current of the tool		N/A
11	**INPUT AND CURRENT**		N/A
	Marked power input or current is at least 110% of measured no-load input or current..... :	See Table 11	N/A
	Tool marked with more than one rated voltage, test made at each rated voltage..... :		N/A
	Tools marked with one or more rated voltage ranges, test made at both the upper and lower limits of the ranges..... :		N/A
	Marking of the rated input is related to the mean value of the relevant voltage range, test is made at a voltage equal to the mean value of that range ..... :		N/A
12	**HEATING**		N/A
12.1	No excessive temperatures attained at rated input or rated current ..... :		N/A
	Temperature rise determined according to Clauses 12.2 to 12.5		N/A
	Test of Clause C.3 at 1,06 times the rated voltage under heated conditions	See Table C.3A	N/A
12.2	Tool is operated at each rated voltage; load conditions as specified in 12.2.1; torque applied is measured and maintained; voltage is then adjusted to 0,94 times and 1,06 times the rated voltage		N/A
	Tool with a rated voltage range is operated at - the lower limit of the rated voltage range; conditions as specified in 12.2.1; torque applied is measured and maintained; voltage is then adjusted to 0,94 times the lower limit of the rated voltage range - the upper limit of the rated voltage range; conditions as specified in 12.2.1; torque applied is measured and maintained; voltage is then adjusted to 1,06 times the upper limit of the rated voltage range		N/A
	Temperatures are measured at the most unfavourable of the voltage settings used		N/A

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Clause	Requirement + Test	Result - Remark	Verdict
	Temperatures measured by means of thermocouples are taken while the tool is operating		N/A
12.2.1	Loading conditions during temperature test ..... :		N/A
	Tool without inherent operating cycle is operated with a torque load to draw rated input or rated current until thermal equilibrium is reached		N/A
	Tool with an inherent operating cycle is operated with a torque load to draw rated input or rated current during each operating cycle; tool was cycled consecutively for 30 min		N/A
12.3.1	Heating elements, if any, are operated under the conditions specified in Clause 11 of IEC 60335-1:2010; tool was operated at 1,06 times the rated voltage		N/A
12.3.2	Tool provided with automatic cord reel, one third of the total length of the cord was unreeled		N/A
12.3.2	Temperature rise was determined near to the hub of the reel and between the two outermost layers of the cord on the reel		N/A
	Cord storage devices, other than automatic cord reels, intended to accommodate the supply cord partially while the tool is in operation, 50 cm of the cord is unwound		N/A
	The temperature rise of the stored part of the cord is determined at the most unfavourable place.		N/A
12.4	Temperature rises, other than those of windings, determined using thermocouples chosen and positioned to have the minimum effect on the temperature of the part tested		N/A
	Temperature rise of electrical insulation, other than windings, measured on surface of insulation		N/A
	When possible, temperature rises of windings determined by resistance method		N/A
	For handles, knobs, grips and the like, all parts considered which are gripped in normal use, and, if of insulating material, to those parts in contact with hot metal		N/A
12.5	Temperature rises did not exceed values in Tables 1a and 1b, except as allowed by 12.6		N/A
	Protective devices did not operate		N/A
	Sealing compounds did not flow		N/A
12.6	When winding temperatures exceeded values in Table 1, three additional samples successfully subjected to following tests:		N/A
	a) Heat treatment for 240 h at the specified cabinet temperature (°C):..... :		N/A
	b) No interturn short circuit after oven treatment		N/A

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Clause	Requirement + Test	Result - Remark	Verdict
	c) Humidity treatment in accordance with 14.1		N/A
	d) Tests of Annex D..... :	See Table D.2	N/A
13	**RESISTANCE TO HEAT AND FIRE**		P
13.1	Relevant parts sufficiently resistant to distortion due to heat		P
	Parts of thermoplastic material: - provided as enclosure to comply with Clause 9, - supporting current carrying parts, - providing supplementary or reinforced insulation, sufficiently resistant to distortion due to heat		P
	Relevant parts subjected to ball-pressure test acc. to IEC 60695-10-2	See Table 13.1	P
13.2	Part of non-metallic material, except as listed in this clause, resistant to ignition and spread of fire		P
	Parts of non-metallic material other than - material classified at least HB40 per IEC 60695-11-10:2013, provided test sample not thicker than relevant part, - material with a glow wire ignition temperature of at least 575 °C per IEC 60695-2-13:2010, provided that the test sample was no thicker than the relevant part, comply with glow-wire test of IEC 60695-1 at 550 °C	See Table 13.2	P
	Soft, foamy, and similar materials which cannot be subjected to glow wire test complies with ISO 9772:2012 for category HBF material with test sample not thicker than relevant part		P
14	**MOISTURE RESISTANCE**		P
14.1	Tools are proof against likely humid conditions		P
	Tool subjected to humidity treatment test for 48 h		P
	Relative humidity (93 ± 2) % ..... :		P
	Temperature (20…30 °C) maintained at ± 1K..... :		P
	Samples pre-conditioned to between t and t + 4 °C:		P
	No excessive leakage after humidity treatment ..... :	See Table C.2A	P
	No flashover or breakdown occurred during test of Annex D after humidity treatment ..... :	See Table D.2	P
	No flashover or breakdown occurred during additional test of D.2 between accessible metal parts and supply cord wrapped with metal foil..... :	See Table D.2	P
14.2	Degree of protection for tool enclosure according to tool classification (IP Code)..... :		N/A
14.2.1	Tool not connected to the supply and turned continuously through most unfavourable positions		P

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Clause	Requirement + Test	Result - Remark	Verdict
	Removable parts are removed and subjected to the relevant treatment with the main part. ..... :		P
14.2.2	Tool rated IPX1 through IPX7 subjected to applicable tests of IEC 60529:2013		N/A
	For IPX7 test, tool immersed in water containing 1,0 % NaCl		N/A
	Tool withstood electric strength test of Annex D after moisture treatment	See Table D.2	N/A
	No trace of water on insulation causing reduction of creepage and clearance below values in 28.1		N/A
14.3	No increased risk of electrical shock from liquid systems or spillage of liquid		N/A
	Residual current device is disabled		N/A
	Removable parts, except those fulfilling the test of 21.22., are removed		N/A
	Tool prepared as described in 8.14.2		N/A
	Liquid container filled, then 15% or 0,25 l added ..... :		N/A
	Detachable liquid container mounted and dismounted 10 times		N/A
	No excessive leakage ..... :	See Table C.3B	N/A
	No flashover or breakdown occurred during test of D.2 between live parts and accessible parts after drying for 24 h at ambient temperature ..... :	See Table D.2A	N/A
14.4	No increased risk of electrical shock from liquid systems under pressure during operation		N/A
	Residual current device is disabled		N/A
	Liquid system is subject to a hydrostatic pressure equal to twice the pressure stated in 8.14.2 d) 1) is applied for 1 h with 1,0 % NaCl solution		N/A
	Tool did not exceed maximum allowable leakage current during pressure application	See Table C.2B	N/A
	No flashover or breakdown occurred during test of D.2 between live parts and accessible parts after drying for 24 h at ambient temperature ..... :	See Table D.2	N/A
14.5	Residual current devices complied with IEC 61540:1999 and met requirements a) to c)		N/A
	a) RCD disconnected only both mains conductors when leakage exceeded 10 mA with a maximum response of 300 ms		N/A
	Test conducted according to 9.9.2 of IEC 61540:1999, and earthing conductor stayed connected		N/A
	b) RCD operated correctly for all 50 cycles		N/A

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Clause	Requirement + Test	Result - Remark	Verdict
	c) RCD cannot be removed during use or routine normal maintenance (i.e., residual current device fixed to tool or power supply cord connected to tool)		N/A
	RCD fitted in supply cord provided with Type Y or Z attachment for connection to supply cord and interconnection cord		N/A
15	**RESISTANCE TO RUSTING**		P
15.1	Ferrous parts adequately protected against rusting		P
	Parts used to conduct electricity subjected to test . :		—
	Mechanical parts mechanical parts specified in the relevant part of IEC 62841-2, IEC 62841-3 or IEC 62841-4 subjected to test..... :		—
	All grease removed from the parts to be tested by immersing them in a degreasing agent for 10 min		P
	Parts were immersed for 10 min in a 10 % solution of ammonium chloride in water at (20 ± 5) °C		P
	Without drying, all drops shaken off, and parts placed for 10 min in a box containing air saturated with moisture at (20 ± 5) °C		P
	After parts dried for 10 min in a heating cabinet at (100 ± 5) °C, no evidence of rust on surfaces		P
	Small helical springs and the like and parts exposed to abrasion covered by a layer of grease		P
16	**OVERLOAD PROTECTION OF TRANSFORMERS AND ASSOCIATED CIRCUITS**		N/A
16.1	No excessive temperatures occurred during short circuit in transformer or circuits associated with it for a tool supplied from a transformer ..... :	See Table 16.1	N/A
	Insulation on conductors of SELV circuits was within 15 K of Table 1		N/A
	Temperature of transformer windings did not exceed values in Table 3		N/A
	Transformer complies with IEC 61558-1		N/A
	Power limited by (short-circuit protective device)... :		—
17	**ENDURANCE**		N/A
17.1	Construction prevents electrical or mechanical failures that might impair compliance with this standard.		N/A
	Insulation not damaged		N/A
	Connections did not work loose		N/A
	Overload protection devices did not activate		N/A

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Clause	Requirement + Test	Result - Remark	Verdict
	No flashover or breakdown occurred during test of Annex D, test voltages reduced to 75 per cent, after tests of 17.2 and 17.3	See Table D.2	N/A
17.2	No load intermittent operation (2 x 24 h) for hand-held tools		N/A
	No load intermittent operation (2 x 12 h) for transportable tools		N/A
	Test voltage at each operation (V)..... :		—
	Rate of operation (100s “on”, 20s “off”) ..... :		—
	Three test positions selected for hand-held tools... :		—
	Normal working position(s) for transportable tools. :		—
	Operation time for each position..... :		—
	Servicing of carbon brushes and lubricant ..... :		N/A
	Replacement of parts due to mechanical failure .... :		N/A
	Forced cooling or rest periods if temperature exceeded values in Table 1 ..... :		N/A
	No operation of overload protection devices		N/A
17.3	Tools with Centrifugal switches operated for 10,000 cycles		N/A
	Number of operations under normal load..... :		N/A
	Rate of operations (s “on”, s “off”)..... :		N/A
	Test voltage 0.9 x rated Voltage (V) ..... :		N/A
18	**ABNORMAL OPERATION**		P
18.1	Risk of fire and mechanical damage impairing - safety and - the protection against electric shock as a result of abnormal operation is obviated as far as is practicable.		P
18.1.1	Tool did not emit flames or molten metal		P
	Compliance with Clause 9 maintained		P
	No flashover or breakdown occurred during test of Annex D between live parts and accessible parts after tests of clause 18	See Table D.2	N/A
	Tool still operable and continues to comply with 19.1 but without repeating the tests of Clause 20		N/A
18.2	Fuses, thermal cut-outs, overcurrent protection devices used to provide the necessary protection		N/A
	Electronic circuits relied upon for protection evaluated for this safety critical function as in clause 18.8.		N/A

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Clause	Requirement + Test	Result - Remark	Verdict
18.3	Tool with series motor operated without accessories at no load for 1 min at 1,3 times rated voltage, or upper limit of voltage range (V) ..... :		—
	No parts were ejected from the tool		N/A
	Speed limiting device operated		—
18.4	Tools with multiphase motor tested, started from cold, with one phase disconnected, and under the torque produced while operated at rated voltage or the mean value of the rated voltage range with rated input or rated current - for 30 s tests for tool kept switched on by hand or continuously loaded by hand - for 5 min test for other tools ..... :		N/A
	30 s tests for tool kept switched on by hand or continuously loaded by hand		N/A
	5 min test for other tools..... :		N/A
	After the test, or at the instant of operation of fuses, thermal cut-outs, motor protection devices and the like, the temperature of the windings complied with the limits in Table 3		N/A
	Max winding temperature recorded (°C) ..... :		—
18.5	Class I tool with class II construction and class II tool subjected to running overload conditions		N/A
	Tools with series motor, test of 18.5.1		N/A
	Class I tool with class II armature test of 18.5.2 instead of 18.5.1		N/A
	Tool with electronically commutated stator windings, test 18.5.4		N/A
	Tool with other motor, test of 18.5.3		N/A
	Lawn and garden machinery, test as specified in relevant part of IEC 62841-4..... :		N/A
18.5.1	All fuses, thermal cut-outs, overload protectors and the like that are accessible or can be reset by the user without the aid of a tool and any self-resetting protective devices were shorted		N/A
	Functions of electronic circuits that prevent the tool from operating at 160 % rated current disabled .... :		N/A
	Functions of electronic circuits that prevent the tool from operating at 160 % rated evaluated as safety critical functions as in 18.8. ..... :		N/A
	Test circuit minimum 12 kVA ..... :		N/A
	Leakage current between live parts and accessible parts measured as in Clause C.3 did not exceed 2 mA throughout the test and until stabilization afterwards ..... :	See Table C.3C	N/A
	Tool operated for 15 min, or until the tool open-circuited, or flame appeared..... :		N/A

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Clause	Requirement + Test	Result - Remark	Verdict
	160% rated test current (A) ..... :		—
	Tool operated at rated voltage (V) ..... :		—
	Overload condition existed for (_min, _sec) ..... :		—
	Condition continued until the tool open-circuited, or flame appeared or 15 minutes expired ..... :		N/A
	Elements that opened in case an open circuit occurred ..... :		N/A
	When flames appeared, extinguished by CO_2 extinguisher		N/A
	Tool did not operate after 15 min, cooled to ambient temperature and subjected to test of D.2 at 1500 V between live parts and accessible parts	See Table D.2	N/A
	Tool still operated after 15 min, cooled to ambient temperature and subjected to test of D.2 at 2500 V between live parts and accessible parts	See Table D.2	N/A
	Tool permanently open-circuited due to over temperature condition (except opening of a motor winding), test repeated.		N/A
	Non-self-resetting thermal limit function of an electronic circuit bypassed or evaluated as a safety critical function in 18.8.		N/A
	Tool permanently open-circuited for reasons other than above, the cause is determined and bypassed in a new sample, test repeated		N/A
18.5.2	Test circuit minimum 12 kVA applied to armature.. :		N/A
	Leakage current between commutator segments and the armature shaft measured did not exceed 2 mA throughout the test and until stabilization afterwards ..... :		N/A
	1,06 times rated voltage (V) applied between opposite commutator segments ..... :		—
	160% rated test current (A) ..... :		—
	Current applied for 15 min, or until the armature open-circuited, or flame appeared ..... :		N/A
	When flames appeared, extinguished by CO_2 extinguisher		N/A
	Armature cooled to ambient temperature and subjected to test of D.2 at 1500 V between commutator segments and the armature shaft	See Table D.2	N/A
18.5.3	Test circuit minimum 12 kVA ..... :		N/A
	Tool stalled, capacitors in circuit of auxiliary windings are open-circuited		N/A
	Test repeated with capacitors short-circuited one at a time unless they are of class P2 of IEC 60252-1		N/A
	Operated at rated voltage (V) ..... :		—

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Clause	Requirement + Test	Result - Remark	Verdict
	Test duration (min, s) ..... :		—
	Temperature of the windings did not exceed the relevant value specified in Table 3		N/A
	Conditions of 18.1.1 fulfilled		N/A
18.5.4	Motors with electronically commutated stator windings, all possible static faults of the outputs of the motor drive circuitry considered		N/A
	Protective function prevent these faults evaluated as an SCF according to 18.8 with minimum PL = a.... :		N/A
	All fuses, thermal cut-outs, overload protectors and the like that are accessible or can be reset by the user without the aid of a tool and any self-resetting protective devices were shorted		N/A
	Leakage current between live parts and accessible parts measured as in Clause C.3 did not exceed 2 mA throughout the test and until stabilization afterwards ..... :	See Table C.3D	N/A
	Voltage applied for 15 min, or until the armature open-circuited, or flame appeared ..... :		N/A
	Source voltage of the motor drive circuitry ..... :		N/A
	When flames appeared, extinguished by CO_2 extinguisher		N/A
	Any motor windings open-circuited after 15 min, motor cooled to ambient temperature and subjected to test of D.2 at 1500 V between live parts and accessible parts	See Table D.2	N/A
	No motor windings open-circuited after 15 min, motor cooled to ambient temperature and subjected to test of D.2 at 2500 V between live parts and accessible parts	See Table D.2	N/A
18.6	No hazards from electric shock, fire or accessible moving parts occurred under fault conditions of 18.6.1		N/A
	Tool operated at rated voltage (V) ..... :		—
	No charring or burning of the gauze or tissue paper occurred		N/A
	Protection against electric shock as in Clause 9 maintained		N/A
	Protection against accessibility to moving parts as in 19.1 maintained		N/A
	Evaluation not performed for low power circuits as in Annex H if no SCF can be lost..... :		N/A
	Circuit encapsulated with an insulating material with a minimum thickness of 0,5 mm and no SCF can be lost, circuit evaluated by open-circuiting and short-circuiting within the encapsulated circuit..... :		N/A

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Clause	Requirement + Test	Result - Remark	Verdict
	Fuses, thermal cut-outs, thermal links, temperature limiters, electronic devices or any components or conductors operated, and		N/A
	– test repeated twice, using two more samples; or		N/A
	– tool withstands test of 18.6.1 with the fuse, thermal cut-out or thermal link bridged; or		N/A
	–miniature fuse link complying with IEC 60127 operates and tool withstands test of 18.6.2		N/A
	Tool withstood the particular test as a conductor of a PCB open-circuited, and		N/A
	– creepage or clearances between live parts and accessible metal parts not reduced below values in 28 due to loosened conductors, and		N/A
	– tool withstood repeated tests with the open-circuited conductor bridged, or		N/A
	– test repeated twice, using two more samples		**N/A**
18.6.1	Fault conditions a) to f) conducted as applicable	See Table 18.6.1	N/A
18.6.2	Tests repeated with fuse-link replaced by an ammeter when during fault conditions of 18.6.1, safety of the tool depended on operation of a miniature fuse-link complying with IEC 60127-3,		N/A
	– Circuit not considered to be adequately protected when current measured was ≤ 2.1 times the rated current of fuse-link, and test conducted with fuse-link short-circuited (A) ..... :		N/A
	– Circuit considered adequately protected when current measured was ≥ 2.75 times the rated current of fuse-link (A)..... :		N/A
	– Fuse-link short-circuited when current measured was 2.1-2.75 times the rated current of fuse-link, and test conducted as follows (A) ..... :		N/A
18.7	Switches and devices for motor reversal withstood stresses occurring when rotation reversed 25 times under running conditions at rated voltage at no-load (V)..... :		N/A
18.8	**Electronic circuits providing safety critical functions (SCF)**		—
18.8.1	Electronic circuits providing SCF are reliable and not susceptible to loss of SCF due to electro-magnetic environmental stresses		N/A
	No SCF lost after tests of 18.8.2 to 18.8.6 for circuits with no internal clock frequency or oscillator frequency > 15 MHz		N/A
	No SCF lost after tests of 18.8.2 to 18.8.7 for other electronic circuits		N/A
	Test voltage was rated voltage or the mean value of the rated voltage range ..... :		N/A

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Clause	Requirement + Test	Result - Remark	Verdict
	Difference between upper and lower limit of rated voltage range > 20 % of its mean value, test at both upper and lower limits of the rated voltage range .. :		N/A
	After evaluation using 18.6.1, no loss of any SCF or tool in a safe state under any present fault condition.		N/A
	Concept of 18.6.1 not appropriate, reliability evaluated using ISO 13849-1.		N/A
	Required performance levels ..... :	See Table 18.8.1A	N/A
	If only $MTTF_d$ is applied to achieve the required PL: $MTTF_d$ is 5/20/50 years for PL = a/b/c		N/A
	Software used in circuits of programmable devices whose failure would create loss of safety critical function, complied with software class B requirements as in H.11.12.3 of IEC 60730-1:2010	See Table 18.8.1B	N/A
	In the case where software class B is realized by single channel with periodic self-test, an acceptable period is regarded as either after each activation of the power switch or a maximum of 5 min.		N/A
	Class B realized by single channel, periodic self-test either after each activation of the power switch or at least every maximum 5 min		N/A
	H.11.12.3.4.1 applicable for SCF with a PL ≥ c		N/A
18.8.2	Electrostatic discharges as in IEC 61000-4-2:2008 applied to tool, test level 4 used for air discharge and test level 3 for contact discharge, ten / ten discharges having a positive / negative polarity applied		N/A
18.8.3	Fast transient bursts as in IEC 61000-4-4:2012 applied to tool, test level 3 used. Repetition frequency 5 kHz for 2 min / 2 min with a positive / negative polarity		N/A
18.8.4	Voltage surges as in IEC 61000-4-5:2005 applied to power supply terminals, five positive impulses and five negative impulses applied at the selected points		N/A
	Test level 3 applied for line-to-line coupling mode, a generator with 2 Ω source impedance being		N/A
	Test level 4 applied for line-to-earth coupling mode, a generator with 12 Ω source impedance being		N/A
	Tools has surge arresters incorporating spark gaps, test was repeated at 95 % of the flashover voltage		N/A
18.8.5	Injected currents as in IEC 61000-4-6:2008 applied to tool, test level 3 applicable, all frequencies between 0,15 MHz to 230 MHz covered		N/A
18.8.6	Class 3 voltage dips and interruptions in accordance with IEC 61000-4-11:2004 applied to tool		N/A
	Values of Tables 1 and 2 of IEC 61000-4-11:2004 were applied at zero crossing of the supply voltage		N/A

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Clause	Requirement + Test	Result - Remark	Verdict
18.8.7	Radiated fields in accordance with IEC 61000-4-3:2010 applied to tool, test level 3 applicable		N/A
	Frequency ranges 80 MHz to 1 000 MHz tested		N/A
19	**MECHANICAL HAZARDS**		P
19.1	Adequate protection against injury provided against moving and other dangerous parts		P
	Protective enclosures, covers, and the like have adequate mechanical strength and cannot be removed without the aid of a tool		P
	Adjustable guard used as protection of the working element has easily accessible means of accurate adjustment		P
	No dangers from adjusting the guards		P
	No contact with dangerous moving parts using probe B of IEC 61032:1997, test force ≤ 5N		P
	Any soft materials removed prior to the test		P
19.2	No hazardous ragged or sharp edges, other than necessary for the functioning of the tool		P
19.3	No contact with dangerous moving parts through dust collection openings, using probe B of IEC 61032:1997, test force ≤5N		P
19.4	Hand-held tool has at least one handle or grasping surface for safe handling during use		P
	Transportable tools provided with at least one handle, grasping surface or the like for safe transportation		P
	Lawn and garden machinery has adequate grasping surfaces for safe handling during use		P
19.5	Tool allows visual check of the contact of cutting tool with workpiece		P
19.6	Marking with rated no-load speed required, measured no-load speed of the spindle did not exceed 110 % of the rated no-load speed		N/A
19.7	Transportable tool or lawn and garden machinery intended to be used on a surface such as the floor or a table has adequate stability		N/A
	10° tilting test, tool or machinery did not tip over		N/A
	Tested with doors open and closed		N/A
	Filled with most unfavourable quantity of water or the recommended liquid		N/A
19.8	Transportable tool provided with wheels identified in the relevant part of IEC 62841-3 has adequate stability during transportation		N/A
	10° tilting test, tool did not tip over		N/A

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Clause	Requirement + Test	Result - Remark	Verdict
19.9	Fixed guards to be removed to convert the tool or to change the accessory, fastenings remains attached to the guard or to the machinery		N/A
	Fastening not completely removed and considered as still attached		N/A
20	**MECHANICAL STRENGTH**		P
20.1	Adequate mechanical strength to withstand rough handling		P
	No flashover or breakdown occurred during test of Annex D between live parts and accessible parts after tests of clause 20.2-20.4	See Table D.2	P
	No live parts became accessible		P
	No creepage distances or clearances below the values of 28.1		N/A
	Mechanical safety of the tool as required by this standard not impaired		N/A
	Inner cover withstood test after removal of the decorative cover		N/A
20.2	Three blows applied to every weak point of enclosure by spring-operated impact test apparatus in Clause 5 of IEC 60068-2-75:1997		N/A
	Brush cap impact energy (Nm) ..... :		—
	Other part impact energy (Nm) ..... :		—
	Blows applied each point of the enclosure likely to be weak..... :		N/A
	Blows applied to guards, covers, handles, levers, knobs and the like as necessary..... :		N/A
20.3	Test of 20.3.1, 20.3.2 or the relevant part of IEC 62841-4 applied, as applicable		N/A
20.3.1	Hand-held tool withstood impact of 3 varied drops on a concrete surface from 1 m		P
	Separable accessories were not mounted		P
	Any attachments provided as specified in instructions, test repeated with each attachment or combination of attachments mounted to a separate tool sample		P
20.3.2	Transportable tool withstood impact with Ø (50 ± 2) mm, (0,55 ± 0,03) kg steel sphere, travelling vertically by (1,3 ± 0,1) m.		P
	Drop test applied to part of the tool that can be impacted from above		P
	Pendulum test applied to part of the tool that cannot be impacted from above		N/A

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Clause	Requirement + Test	Result - Remark	Verdict
	Guard became disassembled but could be reassembled to function properly.		N/A
	Guard became deformed but could be restored to its original shape		N/A
	Other damage, except to guard, accepted, as tool was incapable of normal operation		N/A
20.4	Adequate mechanical strength of brush holder and their caps		P
	Brush cap removed and replace 10 times applying specified tightening torque		P
	Tightening torque (Nm) ..... :		—
	No damage to brush holders impairing its further use, thread not damaged, cap shows no cracks		P
20.5	Handles and grasping surfaces have adequate mechanical strength to provide insulation between grasping area and output shaft		P
	A separate sample subjected to a single impact from 1m onto a concrete surface on each handle and each recommended grasping surface		P
	No flashover or breakdown occurred during test of D.2 at 1250 V a.c. between handles and grasping surfaces in contact with foil and the output shaft of the tool	See Table D.2	P
21	**CONSTRUCTION**		N/A
21.1	Hazardous accidental changing of settings to suit different voltages or speeds unlikely to occur		N/A
21.2	Accidental changing of settings of control devices unlikely to occur		N/A
21.3	Removal of parts ensuring required degree of protection against moisture not possible without aid of a tool		N/A
21.4	Fixing of handles, knobs and the like, used to indicate position of switches or similar components in a hazardous wrong position, was not possible		N/A
21.5	Replacement of a flexible cable or cord requiring displacement of a switch was possible without subjecting internal wiring to undue stress		N/A
	After repositioning of the switch and before reassembling the tool, verification of correct positioning of internal wiring was possible		N/A
21.6	Wood, cotton, silk, paper and similar fibrous or hygroscopic material not used as insulation, unless impregnated or chemically rendered non-fibrous		N/A
21.7	Ordinary driving belts not relied upon to provide required insulation		N/A

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Clause	Requirement + Test	Result - Remark	Verdict
	Special belt design employed to allow use as electrical insulation		N/A
21.8	Insulating barriers of Class II tools, and parts of Class II tools serving as supplementary or reinforced insulation are:		N/A
	- fixed such that they cannot be removed without being seriously damaged; or		N/A
	- so designed that they cannot be replaced in an incorrect position, and when omitted, the tool will be inoperable or manifestly incomplete		N/A
21.9	Inner conductors of a flexible cable or cord are used as wiring within class II construction and insulated from accessible metal parts by:		N/A
	- the sheath of the supply cord itself, this sheath not being exposed to undue thermal stress, clamping against accessible metal or other mechanical stress that could cause damage to it; or		N/A
	- a sleeve, tubing or barrier complying with the requirements of supplementary insulation.		N/A
21.10	Air-intake of motor enclosures not excessively large		N/A
	6 mm steel ball test applied to air-intake openings other than those adjacent to fan		N/A
21.11	No hazards from parts of Class I tool such as wire, screw, nut, washer or spring becoming loose or falling out of position, and accessible metal not made live		N/A
	Clearance and creepage distances of Class II tool or class II construction not reduced to less than 50% of values shown specified in 28.1		N/A
	Class II tool or Class II construction, other than those of the all-insulated type, provided with an insulating barrier between accessible metal and motor parts and other live parts		N/A
	Class I tool with adequately fixed parts, barriers, and sufficiently large creepage and clearances		N/A
	All wires secured in place independent of terminal connection or solder		N/A
21.12	Supplementary and reinforced insulation not impaired by deposition of dirt, or dust resulting from wear of parts within the tool to the extent that creepage and clearances would be reduced		N/A
	Ceramic material not tightly sintered and similar materials, and beads alone, not used as supplementary or reinforced insulation		N/A
	Parts of Elastomer, natural or synthetic rubber used as supplementary insulation are resistant to aging		N/A

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Clause	Requirement + Test	Result - Remark	Verdict
	Rubber parts so arranged and dimensioned that creepage distances not reduced below values in 28.1, even when cracks occurred		N/A
	Insulated material for embedded heating conductors serves only as basic insulation		N/A
	Ageing test for Elastomer and rubber parts for 70 h at 100±2°C		N/A
	No flashover or breakdown occurred during test of D.2, test voltages reduced to 75 per cent..... :	See Table D.2	N/A
	Rubber parts tested..... :		—
	Immersion test for ceramic material on tight sintering in specified fuchsine solution under no less than 15 MPa		N/A
	Test pressure applied (MPa)..... :		N/A
	Test duration (h) ..... :		N/A
	After the test, freshly broken surfaces did not show any trace of dye visible with normal vision		N/A
	Ceramic parts tested ..... :		—
21.13	Internal wiring, windings, commutators, slip rings and the like, and insulation in general, not exposed to oil, grease, and similar substances		N/A
	Adequate insulation properties of oil, grease, and similar substances used for lubrication of gears and the like with no effect on insulation		N/A
21.14	No access to brushes without aid of a tool		N/A
	When tightening screw-type brush-caps, two surfaces clamped together		N/A
	Locking device retaining brushes in position do not depend upon brush spring tension		N/A
	Screw-type brush-caps accessible from the outside of the tool made of or covered with insulating material of adequate strength, and not projecting beyond surrounding surface of the tool		N/A
21.15	Tool employing a liquid system protects the user against increased risk of shock due to presence of liquid under normal use and faults of liquid system		N/A
	Tools employing liquid system constructed as Class III tools, or		N/A
	- class I or II and provided with a residual current device, and complying with 14.3-14.5, or		N/A
	- class I or class II and designed for use in combination with an isolating transformer and complying with 14.3 and 14.4		N/A

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Clause	Requirement + Test	Result - Remark	Verdict
21.16	Tool with compartment accessible without the aid of a tool and likely to be cleaned in normal use, the electrical connections are not subject to pulling during cleaning		N/A
21.17	Tool is fitted with a power switch to control the motor		N/A
	Switch actuator easily visible and accessible		N/A
21.17.1	For tools incorporating a switch with a lock-off device, and switch trigger is operated by squeezing action closing the fingers towards the palm of the hand, lock-off system designed to ensure sufficient durability against abuse and environmental conditions to prevent start by the switch trigger alone		N/A
21.17.1.1	Relevant tool housing is kept for 1 h in a heating cabinet at 80 °C ..... :		N/A
21.17.1.2	Additional test of 21.17.1.2 for lock-off devices that are self-restoring to the lock-off position		N/A
	Number of cycles as per 23.1.10.2 ..... :		N/A
21.17.1.3	Push force of Table 7 applied to most unfavourable point of the switch actuating member ..... :		N/A
	The switch did not actuate		N/A
	The switch and its lock-off system operated as designed after the applied force was terminated		N/A
21.18	Requirements of 21.18.1, 21.18.2 or the relevant part of IEC 62841-4 observed, as applicable		N/A
21.18.1	Hand-held tool fitted with momentary power switch, unless without a relevant part of IEC 62841-2 and without a substantial risk from continued operation		N/A
	Switch can be switched on and off by the user without releasing any of the required handle(s) or grasping surface(s)		N/A
21.18.1.1	A momentary switch locking in “on” position unlocks automatically upon a single actuation motion without releasing the grasp on the tool		N/A
	More than one switch, the lock-on switch(es), if any, is (are) within the grasping zone necessary to control the tool		N/A
	Any one of these switches automatically unlocks or makes ineffective all remaining lock-on devices with a single actuation motion without releasing the grasp on the tool		N/A
	Switch cannot be locked in “on” position when a risk with continued operation is defined by the relevant part of IEC 62841-2		N/A
21.18.1.2	Power switch triggers and lock-off devices so located, designed or guarded that inadvertent operation is unlikely to occur		N/A

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Clause	Requirement + Test	Result - Remark	Verdict
	Tool did not start when 100 mm sphere is applied to the power switch, or		N/A
	Two separate and dissimilar actions necessary before the motor is switched		N/A
21.18.2	Transportable tool fitted with power switch easily actuated “on” or “off” without any reasonably foreseeable hazard		N/A
21.18.2.1	Power switch in transportable tools is of momentary type, or		N/A
	Voltage recovery following an interruption of the supply gives rise to a hazard		N/A
	Relevant part of IEC 62841-3 ..... :		—
21.18.2.2	“On”/”off” control capable of being turned off by the operator with a single straight-line motion		N/A
	Flap cover covers the stop button so that pushing the flap actuates the stop		N/A
21.18.2.3	Power switch so located, designed or guarded that unintentional movement to the “on” position is unlikely		N/A
	Tool did not start when 100 mm sphere is applied to the power switch, or		N/A
	Two separate and dissimilar actions necessary before the motor is switched		N/A
21.18.2.4	Push-pull switch is turned off by an inward push		N/A
21.19	Protection against electric shock not affected when screws removed during user maintenance are incorrectly replaced during reassembly		N/A
	Creepage and clearances between live parts and accessible metal parts not reduced below values in 28.1 when screws are installed at improper screw locations		N/A
21.20	Tool marked with the first numeral of IP system complies with IEC 60529:2013 ..... :		N/A
21.21	No risk of electrical shock from charged capacitors when touching pins of the plug		N/A
	Max. voltage measured between pins of the plug is ≤ 34 V after 1 s after each disconnection (V) ..... :		N/A
	Capacitors rated ≤ 0.1 μF		N/A
	Capacitors complying with the requirements for protective impedance specified in 9.2 and 21.34		N/A
21.22	Non-detachable protective parts either removable with the aid of a tool or reliably fixed		N/A
	Snap-in devices have an obvious locked position and have fixing properties that do not deteriorate		N/A
	Parts disassembled and assembled 10 times prior to test		N/A

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Clause	Requirement + Test	Result - Remark	Verdict
	Parts affected by temperature tested immediately after conditions of Clause 12		N/A
	Test applied to all parts likely to be detached, whether or not fixed by screws, rivets, or similar parts		N/A
	Weak areas of the covers or parts subjected during 10 s to - 50 N push force		N/A
	- 50 N pull force if the shape of the part prevents easy slippage of fingertips		N/A
	- 30 N pull force if projection of the gripped part is less than 10 mm in the direction of removal		N/A
	Test fingernail of Fig. 1 inserted in apertures and joints with a force of 10 N and then slid sideways with a force of 10 N		N/A
	Axial pull unlikely, test fingernail of Fig. 1 inserted in apertures and joints with a force of 10 N to enable a force of 30 N for 10 s by means of a loop		N/A
	A torque of 2 Nm applied at the same time as pull or push force on parts 50 mm or smaller and likely to be subjected to twisting		N/A
	A torque of 4 Nm applied at the same time as pull or push force on parts larger than 50 mm and likely to be subjected to twisting		N/A
	Projection was less than 10 mm and required a torque of (Nm), test torque reduced..... :		N/A
	Parts not detached, and remained in locked position		N/A
21.23	Handles, knobs, grips, levers etc., withstood axial force of 30 N for 1 minute		N/A
21.24	Storage hooks and similar devices for flexible cords are smooth and well rounded		N/A
21.25	Current-carrying parts and other parts resistant to corrosion under normal use		N/A
	After tests of Clause 15, no sign of corrosion on relevant parts		N/A
	Stainless steel and similar corrosion-resistant alloys and plated steel considered satisfactory		N/A
21.26	Insulation between parts operating at SELV and other live parts complies with the requirements for double insulation or reinforced insulation		N/A
21.27	Insulation between parts separated by protective impedance comply with requirements for double or reinforced insulation		N/A
21.28	Shafts of operating knobs, handles, levers etc. not live unless their removal does not make the shaft accessible to test probe B of IEC 61032:1997		N/A

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Clause	Requirement + Test	Result - Remark	Verdict
21.29	Handles, levers, and knobs of non-class III tool held or actuated in normal use do not become live during an insulation fault		N/A
	Metallic handles, levers, and knobs with shaft or fixings likely to become live due to basic insulation fault, either adequately covered by insulating material or their accessible parts separated from their shafts or fixings by insulation		N/A
	Exception for handles, levers, and knobs of transportable tools and lawn and garden machinery of class I..... :		N/A
	Covering or insulating material complies with Electric Strength test in D.2 at 1250 V	See Table D.2	N/A
21.30	Tool likely to cut into concealed wiring or own cord, handles and grasping surfaces - made of insulating material, or		N/A
	- metal covered by insulating material, or		N/A
	- their accessible parts are separated by insulating barrier(s) from accessible metal parts that may become live by the output shaft		N/A
	Insulated, stick type, auxiliary handle is provided with a flange ≥ 12 mm high above grasping surface between grasping area and accessible parts that may become live by the output shaft		N/A
	21.30 not applicable as per relevant part of IEC 62841-2, IEC 62841-3 or IEC 62841-4..... :		N/A
21.31	Capacitors in class II tools not connected to accessible metal parts, and their metallic casings are separated from accessible metal parts by supplementary insulation		N/A
	Capacitors tied to accessible metal parts comply with Clauses 9.2 and 21.34		N/A
21.32	Capacitors not connected between contacts of the thermal cut-outs		N/A
21.33	Lamp holders used only for connection of lamps		N/A
21.34	Protective impedance consists of at least two separate components with impedance unlikely to change significantly during lifetime of tool		N/A
	When a component short or open-circuited, values in Clause 9.2 were not exceeded ..... :		N/A
	Resistors comply with 14.1 of IEC 60065:2011 and capacitors comply with 14.2 of IEC 60065:2011 .... :		N/A
	Single Y1 capacitor acc. to IEC 60384-14 used instead of two separate components..... :		N/A
21.35	Tools is identified in the relevant part of IEC 62841-2 or IEC 62841-3 to produce a considerable amount of dust and has either integral dust collection/suction device or dust outlet(s)		N/A

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Clause	Requirement + Test	Result - Remark	Verdict
	Dust discharge directed away from the operator		N/A
	Dust outlet with external suction device(s) does not impede the normal use of the tool		N/A
22	**INTERNAL WIRING**		N/A
22.1	Wireways smooth and free from sharp edges, cooling fins, etc		N/A
	Holes in metal through which insulated wires pass provided with bushings or, except as required by relevant part of IEC 62841-2, IEC 62841-3 or IEC 62841-4, have smooth edges with radius ≥1,5 mm		N/A
	Wiring prevented from coming into contact with moving parts		N/A
22.2	Internal wiring adequately rigid, fixed or insulated such that creepage and clearances cannot be reduced below values in 28.1		N/A
	Sleeving used as supplementary insulation on internal wiring, retained in position by positive means (removable only by breaking or cutting, or clamped at both ends)		N/A
22.3	Use of green or green/yellow conductors for earthing terminals only		N/A
22.4	Aluminium wires not used for internal wiring		N/A
22.5	Stranded conductors with lead-tin soldering are only used with spring terminals with constant contact pressure, except when clamping means pose no risk of bad contact		N/A
22.6	No undue stress to electrical connections and internal conductors from tool parts movable to each other in normal use, during adjustment or user maintenance		N/A
	Flexible metallic tubes do not damage insulation of the conductors contained within them		N/A
	Open-coil springs not used to protect the wiring		N/A
	Adequate additional insulating lining when coiled spring is used		N/A
	Flexing test at a rate of ≤ 6/min, through the largest angle allowed by the construction		N/A
	Number of flexings 10 000 for conductors/ connections flexed during normal use; 2 000 for those flexed during adjustments; 100, for those flexed during user maintenance..... :		—
	Tool withstands test of Annex D between live parts and accessible parts	See Table D.2	N/A
	Live parts not accessible after test		N/A

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Clause	Requirement + Test	Result - Remark	Verdict
23	**COMPONENTS**		N/A
23.1	Components comply with relevant IEC standards	See Table 23.1	N/A
	Batteries are regarded as part of the tool and comply with Annexes K and/or L		N/A
	Components used in accordance with their markings		N/A
	Applied exceptions ..... :		N/A
	Components not previously tested and found to comply with the relevant IEC standard for the number of cycles specified, tested to 23.1.1 … 23.1.11 ..... :		N/A
23.1.1	Capacitors in auxiliary windings of motors marked with their rated voltage and rated capacitance..... :		N/A
23.1.2	Fixed capacitors for radio interference suppression comply with IEC 60384-14		N/A
23.1.3	Small lamp holders similar to E10 lamp holders meet requirements for E10 lamp holders in IEC 60238		N/A
23.1.4	Isolating and safety isolating transformers comply with IEC 61558-1 and IEC 61558-2-4 or IEC 61558-2-6, as applicable		N/A
	Switch mode power supply units and transformers for such units comply with IEC 61558-2-16		N/A
23.1.5	Appliance couplers comply with IEC 60320, or		N/A
	Instructions provided to inform user to connect the tool with non-IEC appliance couplers		N/A
23.1.6	Automatic temperature controls with electromechanical contacts that cycle in normal use have suitable endurance		N/A
	Tests to IEC 60730-1:2010, Cl. 17, conducted under conditions occurring in the tool		N/A
	Type of controls used and number of cycles per Cl. 17 of IEC 60730-1:2010 (cycles) ..... :		N/A
	Automatic controls comply with IEC 60730-1:2010, and are used in accordance with their marking		N/A
	Tests of Clause 17 of IEC 60730-1:2010 were not conducted on automatic controls because tool complies with this standard when protective device short-circuited		N/A
	Thermostats and temperature limiters tested in accordance with a specific exception in Note b) of Table 1 of Clause 12		N/A
23.1.7	Unless otherwise specified, tests on components per other standards conducted separately according to the relevant standard		N/A
	Component, marked and used per its markings		N/A
	Components not mentioned in Table 1 of Clause 12 tested as part of the tool		N/A

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Clause	Requirement + Test	Result - Remark	Verdict
23.1.8	Components not separately tested and found to comply with the component standards as references in 23.1 or components not marked or not used in accordance with their marking, tested in accordance with the referenced relevant standard under the conditions occurring in the tool		N/A
	No IEC standard referenced in 23.1, no additional tests		N/A
23.1.9	Tool operated at 1,1 times rated voltage at no-load, capacitor voltage did not exceed 1.1 times its rated voltage (V)..... :		N/A
23.1.10	Switches constructed to prevent failure that might impair compliance with this standard		N/A
	Switches, separately tested and found to comply with IEC 61058-1:2008, comply with 23.1.10.1		N/A
	Switches, not separately tested and found to comply with IEC 61058-1:2008, or not complying with 23.1.10.1, tested as in 23.1.10.2 to 23.1.10.3		N/A
23.1.10.1	Power switches rated for a voltage and current not less than respective ratings of the tool		N/A
	Power switches rated for a.c. in a.c. tools and d.c. in d.c. tools		N/A
	Electronic power switches are at least classified for Continuous Duty as in IEC 61058-1:2008		N/A
	Switches for motor-operated tools and lawn and garden machinery classified for resistive and motor load as in 7.1.2.2 of IEC 61058-1:2008, if this load occurs in normal use		N/A
	Switches for magnetically driven tools and lawn and garden machinery classified for inductive load as in 7.1.2.8 of IEC 61058-1:2008, if this load occurs in normal use		N/A
	Switches alternatively regarded as switches for a declared specific load as in 7.1.2.5 of IEC 61058-1:2008 and classified based on the load conditions of the tool in normal use		N/A
	Ratings and load classifications for switches other than power switches are based on the conditions encountered in the tool		N/A
	Power switches for hand-held tools classified for min. 50K operating cycles ..... :		N/A
	Power switches for transportable tools and lawn and garden machinery classified for min. 10K operating cycles ..... :		N/A
	Power switches with series electronics also endure 1000 operating cycles, electronics bypassed..... :		N/A

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Clause	Requirement + Test	Result - Remark	Verdict
	Switches other than power switches, if likely to be switched under electrical load, endure 1 000 operating cycles, unless the requirements of this standard are met with the switch short-circuited		N/A
	Exception for switches other than power switches that cannot be operated under electrical load		N/A
	Exception for motor reversing switches		N/A
	Exception for switches other than power switches, classified for 20 mA load as in 7.1.2.6 of IEC 61058-1:2008		N/A
23.1.10.2	Adequate endurance properties of switches		N/A
	Test of 17.2.4.4 of IEC 61058-1:2008 conducted at load specified in 23.1.10.2.1 or 23.1.10.2.2..... :		N/A
	Power switches for hand-held tools tested for 50K cycles.		N/A
	Power switches for transportable tools and lawn and garden machinery tested for 10K cycles		N/A
	Power switch contains mechanical contacts in series with electronic circuitry with one or more SSD and circuitry provides a protective function by reducing the current during switch operation, then test repeated on 3 samples for ≥ 1000 cycles with the electronics bypassed; or		N/A
	Protective function considered SCF and complies with the greater PL levels for power switches in 18.8		N/A
	Switches other than power switches, if likely to be switched while energized, tested for 1000 cycles under load conditions of normal use		N/A
	After tests all switches were able to be turned on and off and complied with the insulating compliance (TE3) of 17.2.5.3 of IEC 61058-1:2008 for basic insulation		N/A
23.1.10.2.1	Power switches for motor-operated tools and lawn and garden machinery classified to 7.1.2.2 of IEC 61058-1:2008 and tested with external load as specified		N/A
	Power switches for magnetically driven tools and lawn and garden machinery classified to 7.1.2.8 of IEC 61058-1:2008 and tested with external load as specified		N/A
	Switches other than power switches, but which would encounter the same load conditions as power switches in normal use, tested as specified		N/A
23.1.10.2.2	For switches tested using the motor or magnetic load encountered in the tool, tested at rated voltage for the required number of cycles; tool is switched on at no-load and switched off at rated current or rated input		N/A

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Clause	Requirement + Test	Result - Remark	Verdict
23.1.10.3	Power switches of motor-operated tools and lawn and garden machinery have adequate breaking capacity		N/A
	Locked-rotor test (TC9) of 17.2.4.9 of IEC 61058-1: 2008 at 6 times I-M or with locked motor, each period ≤ 0,5 s “on” and ≥ 10 s “off”		N/A
	Power switch showed no electrical or mechanical failure after test		N/A
23.1.11	Electronic power switches comply with 18.6 and 18.8		N/A
23.2	Tool not fitted with switches or automatic controls in flexible cords, except for protective devices such as RCDs		N/A
	Tool not fitted with devices causing the protection device in the fixed wiring to operate		N/A
	Tool not fitted with thermal cut-outs which can be reset by a soldering operation		N/A
23.3	Protection devices or circuits that switch off the tool are non-self-resetting where a risk associated with inadvertent starting is specified		N/A
23.4	Plugs and socket-outlets for ELV circuits and those used as terminal devices for heating elements not interchangeable with mains plugs and socket-outlets in IEC 60884, IEC/TR 60083 or IEC 60906-1 or with connectors and appliance inlets complying with IEC 60320-1		N/A
23.5	Motors connected to the supply mains with insulation inadequate for the rated voltage comply with Annex B		N/A
24	**SUPPLY CONNECTION AND EXTERNAL FLEXIBLE CORDS**		**N/A**
24.1	Tool provided with a supply cord ≥1,8 m and with a plug; cord length (m)..... :		N/A
	Tool provided with a supply cord at least 1,8 m long and without a plug; cord length (m) ..... :		N/A
	Information for connection given in the instructions		
	Tool provided with appliance inlet having at least same degree of protection against moisture as required for the tool		N/A
	Tool provided with a supply cord ≥ 0,2 m and ≤ 0,5 m and with a plug or other connector having at least same degree of protection against moisture as required for the tool; cord length (m) ..... :		N/A
	Plugs, connectors and inlets suitable for the ratings of the tool		N/A
24.2	Supply cord assembled to the tool by attachment type (specify X, Y, or Z)..... :		N/A

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Clause	Requirement + Test	Result - Remark	Verdict
	Supply cord with type Z attachment is allowed as per relevant part of IEC 62841-2, IEC 62841-3 or IEC 62841-4..... :		N/A
	Supply cords with type X attachment are specially prepared cords only available from the manufacturer or its service agent ..... :		N/A
	Special cord includes part of the tool		N/A
24.3	Plugs fitted with only one flexible cord		N/A
24.4	Supply cord not lighter than ordinary tough rubber sheathed flexible cord or ordinary PVC sheathed flexible cord		N/A
	PVC cords not used if external metal parts exceed 75 K temperature rise during test of Clause 12		N/A
24.5	Nominal cross-section area of supply cord per Table 8 (mm2)..... :		N/A
24.6	Supply cord of class I tool has green or green/yellow core connected to internal earthing terminal of the tool, and to earthing contact of plug		N/A
24.7	Lead-tin solder not used to consolidate leads under contact pressure, except when clamping means used prevents risk of a bad contact		N/A
	Clamping screws alone not used for securing soldered leads		N/A
24.8	Moulding supply cord to any part has no effect on the insulation of the cord		N/A
24.9	Supply cord protected against damage at its entry by flexible cord guard, or cord inlet, or bushing..... :		N/A
24.10	Cord inlets and bushings shaped to prevent damage to supply cord		N/A
	Cord inlet and bushings reliably fixed and not removable without the aid of a tool		N/A
24.11	In tools other than transportable tools, supply cord being flexed during operation is protected against excessive flexing at its entry ..... :		N/A
	Flexing test performed in apparatus shown in Fig. 2		N/A
	Weight attached to cable or cord (kg)..... :		—
	Oscillating member moved back and forth through an angle of 90° (45° on either side of the vertical) with rate of 60 flexings per minute		N/A
	After 10,000 flexings, sample turned through 90° about the centre of the cord entry		N/A
	Cord guard did not slip out from its location after completion of ten 1 sec lifts over 500 mm		N/A
	After the test, no conductor disconnected from terminal		N/A

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Clause	Requirement + Test	Result - Remark	Verdict
	Number of strands versus number of broken strands of each conductor ≤ 10% ..... :		N/A
24.12	In tools other than transportable tools, supply cord being flexed during operation is protected against excessive bending at its entry..... :		N/A
	Cord guard fixed reliably and projects outside tool for a distance beyond inlet opening of at least 5 times the overall diameter of cord		N/A
	Mass attached to the free end of cord (g)..... :		—
	Curvature of cable or cord is nowhere less than 1,5 times the external diameter of cord		N/A
24.13	Tool provided with cord anchorage to relieve conductors of cord from strain, twisting, and protect them from abrasion.		N/A
	Pushing cord into the tool not possible		N/A
	Pull force was applied 25 times at the force shown in Table 9 (N)..... :		—
	After pull test, cord, unless on an automatic cord reel, subjected to torque in Table 9 for 1 min (Nm) :		
	The cord was not damaged during the tests		N/A
	No appreciable strain at the terminals		N/A
	Cord longitudinal displacement (mm) ..... :		N/A
	No appreciable strain at the connection		N/A
24.14	Cord anchorage either accessible only with the aid of a tool, or the cord can only be fitted using a tool		N/A
24.15	Cord anchorages properly designed and located		N/A
	Cord cannot touch clamping screws of the cord anchorage that not separated from accessible metal parts by supplementary insulation		N/A
	Cord not clamped by metal screw bearing directly on the cord		N/A
	Glands are not used as cord anchorages		N/A
	Class I tool, cord anchorage of insulating material or with insulating lining fulfilling basic insulation, if an insulation fault on the cord could make accessible metal parts live		N/A
	Class I tool, sheath of the cord considered adequate		N/A
	Class II tool, cord anchorage of insulating material or insulated by supplementary insulation (sheath of the cord alone not sufficient)		N/A
24.16	Cord anchorages for type X attachment properly designed and located		N/A
	Cord anchorage allows easy replacement of cord		N/A
	Clear method of relief from strain and prevention of twisting		N/A

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Clause	Requirement + Test	Result - Remark	Verdict
	Screws operated during cord replacement are not used to fix any other part		N/A
	Screws operated during cord replacement are used to fix other parts and, if omitted or incorrectly mounted, make the tool inoperative or clearly incomplete		N/A
	Parts fastened to the cord anchorage by the same screw could not be removed without the aid of a tool		N/A
	Conductors inserted into terminals, terminal screws tightened sufficiently to prevent conductors from easily changing their position, torque set at (Nm) .. :		N/A
24.17	Knots and tying strings for type X attachment are not used		N/A
24.18	For type X attachment, space for supply cord provided inside or as a part of tool		N/A
	- permits verification of correct connection and positioning of conductors		N/A
	- permits covers to be fitted without risk of damage to supply conductors or their insulation		N/A
	- ensures that uninsulated end of conductor, when detached from a terminal, cannot come into contact with accessible metal parts, or terminations are unlikely to slip free of the conductor		N/A
	For pillar terminals (with conductors that are not separately clamped ≤ 30 mm from terminal), and for other terminals with screw clamping, a force of 2 N applied to the wire in any direction and adjacent to the terminal, screw or stud		N/A
	The uninsulated end of the conductor did not come into contact with accessible metal parts		N/A
24.19	Appliance inlet prevents access to live parts during insertion or removal of the connector		N/A
	Easy insertion of connector		N/A
	After insertion of connector, tool not supported by the connector in any position of normal use on a flat horizontal surface		N/A
	Test probe B of IEC 61032:1997 applied to tool inlet other than appliance inlet per IEC 60320		N/A
	Appliance inlet complies with IEC 60320		N/A
24.20	Interconnection cords meet the requirements for the supply cord, exceptions as follows		N/A
	Cross-sectional area is based on maximum current through conductor during test of Clause 12		N/A
	Insulation adequate for conductor’s working voltage		N/A
	Test of 24.11 restricted to range of motion during normal use.		N/A

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Clause	Requirement + Test	Result - Remark	Verdict
24.21	Interconnection cords not detachable without tool if compliance with this standard is impaired when they are disconnected		N/A
25	**TERMINALS FOR EXTERNAL CONDUCTORS**		N/A
25.1	Tool provided with terminals or equally effective devices for connection to external conductors		N/A
	Terminals only accessible with the aid of a tool		N/A
	Screws and nuts allowed to also clamp internal conductors when they are unlikely to be displaced when fitting supply conductors		N/A
	Screws and nuts do not fix other components		N/A
	For tool with type X attachment, soldered connections allowed for connection of external conductors, when soldering alone is not used to maintain conductor in position		N/A
	When provided, barriers prevent creepages and clearances between live parts and other metal parts from being reduced to < 50% of values in 28.1, the conductor can be fixed by soldering alone		N/A
	For type Y and Z attachments, soldered, welded, crimped and similar connections allowed for the connection of external conductors		N/A
	Class II tools, conductor so positioned or fixed that soldering, crimping, or welding alone not relied upon to maintain the conductor in the position		N/A
	Barriers prevent creepages and clearances between live parts and other metal parts from being reduced to < 50% of values in 28.1 for the Class of tool using Type Y or Z attachments		N/A
	Conductors connected by soldering are held in place near termination independent of solder		N/A
	Conductor is “hooked in” before soldering and the hole through which it passes is not too large		N/A
	Terminals of a component built into the tool used to secure external conductors		N/A
	Conductors connected by other means, leads additionally fixed near terminations		N/A
	Stranded conductors secured at insulation and conductor		N/A
25.2	Terminals for supply cords suitable for their purpose		N/A
	Supply cord terminals withstood pull force of 5 N		N/A
25.3	For type X attachment, when clamping means tightened or loosened, terminal did work loose, no stress on internal wiring, and creepage and clearances not reduced below values in 28.1		N/A

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Clause	Requirement + Test	Result - Remark	Verdict
	Test per Clause 9.6, using 2/3 torque of that in Table 4, of IEC 60999-1:1999 (Nm) ..... :		N/A
	Terminals secured by two screws to prevent loosening, or by one screw in a recess, or by other suitable means		N/A
	Correct position of supply terminals maintained by switches and similar devices with recesses and verified after connection of supply cord and repositioning of device		N/A
	Sealing compound without other means of locking not used		N/A
	Self-hardening resins used only on terminals that are not subject to torsion in normal use		N/A
25.4	Type X attachment using terminals to clamp the conductor between metal surfaces do so without damage to conductor after torque test per Cl. 25.3		N/A
25.5	End of conductor inserted in the hole of pillar type terminals is visible, or can pass beyond threaded hole for a distance of half nominal diameter of screw, or 2,5 mm, the greater of the two (mm) ..... :		N/A
25.6	For type X attachment, terminals clearly recognizable and accessible after opening the tool		N/A
	All terminals located behind one cover, or one part of the enclosure		N/A
25.7	For tool with type X attachment, terminal devices located or shielded to prevent a strand of wire from escaping		N/A
	No risk of accidental connection between live parts and accessible metal parts		N/A
	For class II tool, no risk of accidental connection between live parts and metal parts with supplementary insulation only		N/A
	8 mm long free wire of the stranded supply conductor did not touch any accessible metal part		N/A
	8 mm long free wire of the stranded supply conductor did not touch any metal parts with supplementary insulation only		N/A
	8 mm long free wire of stranded conductor connected to an earthing terminal did not touch any live part		N/A
			N/A
26	**PROVISION FOR EARTHING**		
26.1	Accessible metal parts of class I tool permanently connected to an earthing terminal or termination within the tool		N/A
	Accessible metal parts of class I tool permanently connected to the earthing contact of the tool inlet		N/A

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Clause	Requirement + Test	Result - Remark	Verdict
	Printed circuit boards are not used to provide continuity of protective earthing circuit		N/A
	No electrical connection between earthing terminals or contacts and neutral terminal		N/A
	No provision for earthing in Class II and III tools		N/A
	Rotating motor components with metal-to-metal bearing surfaces considered electrically bonded		N/A
	Metal parts behind a decorative cover that do not withstand test of Clause 20 considered accessible metal parts		N/A
26.2	Clamping means of earthing terminals adequately locked against accidental loosening		N/A
	Earthing connections not possible to loosen without the aid of a tool		N/A
	Terminals with screw clamping comply with the relevant requirements of Clause 25, and screwless terminals comply with IEC 60998-2-2		N/A
	For specially prepared cords, terminals comply with IEC 61210 and table 10		N/A
	Screwless terminals tested per IEC 60998-2-2		N/A
26.3	Earth connection of detachable parts was made before the current-carrying connections established when placing the part in position, and the current carrying connections separated before earth connection was broken when removing the part		N/A
	If cord slips out of cord anchorage, current-carrying conductors become taut before earthing conductor		N/A
26.4	No risk of corrosion between metal parts of earthing terminals and copper of earthing conductor		N/A
	Parts transmitting current in case of an insulation fault, other than parts of metal frame or enclosure, are coated or uncoated metal with adequate resistance to corrosion		N/A
	Thickness of electroplated coating (μm)..... :		N/A
	Parts of coated or uncoated metal providing or transmitting contact pressure only, adequately protected against rusting		N/A
	Protection provided against risk of corrosion resulting from contact between copper and aluminium (or aluminium alloy)		N/A
	Parts subjected to a treatment such as chromate conversion coating are used only to provide or transmit contact pressure		N/A
	Thickness of coating of steel measured in accordance with ISO 2178 or ISO 1463 (μm) ..... :		N/A
	Resistance to rusting test..... :	See also 15.1	N/A

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Clause	Requirement + Test	Result - Remark	Verdict
26.5	Resistance of earthing circuit (max. 0.1 Ω) ..... :		N/A
	Test current (A)..... :		—
	Voltage drop between the earthing terminal and accessible metal part (V)..... :		—
27	**SCREWS AND CONNECTIONS**		P
27.1	Fixings and electrical connections (earthing connections included) withstand mechanical stresses occurring in normal use		P
	Screws not made of soft metal such as zinc or aluminium		P
	Diameter of screws of insulating material not used for electrical or earthing connection, diameter (mm). ..... :		N/A
	Screws transmitting electrical contact pressure screw into metal		N/A
	Screws of insulating material not used if their replacement by a metal screw could impair supplementary or reinforced insulation		N/A
	Screws removed when replacing the supply cord with type X attachment, or during maintenance, are not of insulating material where their replacement by a metal screw could impair basic insulation		N/A
	Screws and nuts tightened and loosened 10 times for screw engaged with a thread of insulating material		N/A
	Nuts and other screws tightened and loosened five times		N/A
	Screws engaging with a thread of insulating material completely removed and reinserted each time		N/A
	When testing terminal screws and nuts, a flexible conductor of the largest cross-sectional area per Clause 24.5 placed, and each time repositioned, in the terminal (mm^2) ..... :		N/A
	Test using a suitable test screwdriver, spanner or key, torque as in Table 11 and the relevant column		N/A
	Column I for metal screw without head, flush with surface (Nm) ..... :		N/A
	Column II for other metal screws and nuts (Nm).... :		N/A
	Column II for screws of insulating material, having a hexagonal head with a width across flats exceeding overall thread diameter (Nm) ..... :		N/A
	Column II for screws of insulating material, having a cylindrical head and a key socket with a width across corners exceeding overall thread diameter (Nm) ... :		N/A

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Clause	Requirement + Test	Result - Remark	Verdict
	Column II for screws of insulating material, with a head having a slot or cross-slots longer than 1,5 times the overall thread diameter (Nm) ..... :		N/A
	Column III applied to other screws of insulating material (Nm) ..... :		N/A
	No damage impairing further use of fixing or electrical connections		N/A
27.2	Contact pressure not transmitted through insulating material other than ceramic, unless compensated for shrinkage or distortion		N/A
27.3	Space-threaded screws not used for connection of current-carrying parts, unless direct clamping and suitable locking provided		N/A
	No thread-cutting screws used for connection of current-carrying parts		N/A
	Use of two space-threaded or thread-cutting screws in earthing circuits		N/A
27.4	Screws making both mechanical and electrical connections are locked against loosening		N/A
	Rivets for current-carrying connections subjected to torsion in normal use locked against loosening		N/A
27.5	Screwless connectors not intended for disconnection in normal use prevent disconnection in normal use		N/A
	Connectors withstood 5 N pull through the wire		N/A
	Neither the connector nor the wire became disconnected		N/A
	Directions of the application and exit of the wire not in line, force applied in both directions, one at a time		N/A
	Connectors fulfilled relevant IEC standards and were considered to fulfil requirements of 27.5.		N/A
27.5.1	Conductors secured by more than one means, unless their detachment does not impair safety		N/A
	Only one means of securing, test with detached conductors		N/A
	Clearances not reduced below 50 % of values in 28.1		N/A
28	**CREEPAGE DISTANCES, CLEARANCES AND DISTANCES THROUGH INSULATION**		N/A
28.1	Creepage and clearances not less than the values in Table 12, except for cross-over points of motor windings ..... :	See Table 28.1	N/A
	When a resonance voltage occurs, creepage and clearance are not less than specified for the voltage imposed by the resonance; these values increased by 4 mm in case of reinforced insulation		N/A

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Clause	Requirement + Test	Result - Remark	Verdict
	Creepage and clearances for a tool with an appliance inlet measured with an appropriate connector inserted		N/A
	Creepage and clearances on a tool with other attachment measured on the tool as delivered		N/A
	Measurements on tool with belt made with the belt in place and belt tension adjusted to the most unfavourable position within its adjustment range		N/A
	Measurements repeated with the belt removed		N/A
	Movable parts placed in the most unfavourable position		N/A
	Nuts and screws with non-circular heads tightened in the most unfavourable position		N/A
	Clearances between terminals and accessible metal parts also measured with screws and nuts unscrewed as far as possible and they were not less than 50% of Table 12 ..... :	See Table 28.1	N/A
	Distances through slots or openings in external parts of insulating material measured to metal foil in contact with accessible surface with the foil pushed into corners using test probe B of IEC 61032:1997 :	See Table 28.1	N/A
	2 N force applied to internal wiring, bare conductors and uninsulated capillary tubes of thermostats and similar devices during measurement		N/A
	30 N force applied to enclosure		N/A
	Measurements made according to Annex A	See Table 28.1	N/A
	Creepage and clearances on a tool having parts with double insulation and no metal between basic insulation and supplementary insulation		N/A
	PWB with peak voltage stresses ≤ 150 V per mm between parts of different potential provided with a min. distance of 0.2 mm, when protected against deposition of dirt	See Table 28.1	N/A
	-PWB with 100 V per mm provided with a min. distance of 0.5 mm, when not protected against deposition of dirt	See Table 28.1	N/A
	Values of the table applied when limits mentioned above resulted in higher values than in the table	See Table 28.1	N/A
	Distances reduced further since the tool complied with the requirements of Clause 18 distances short-circuited one at a time ..... :	See Table 28.1	N/A
	Creepage and clearances within optocouplers not measured when individual insulation adequately sealed, with air excluded between material layers		N/A
	For live parts of different polarity separated by basic insulation only, creepage and clearances reduced as tool complied with Clause 18 when creepage and clearances short-circuited ..... :	See Table 28.1	N/A

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Clause	Requirement + Test	Result - Remark	Verdict
28.2	Distance through insulation between metal parts was ≥1.0 mm for working voltages ≤130 V when separated by supplementary insulation	See Table 28.2	N/A
	Distance through insulation between metal parts was ≥1.5 mm for working voltages ≤130 V when separated by reinforced insulation	See Table 28.2	N/A
	Distance through insulation between metal parts was ≥1.0 mm for working voltages > 130V ≤ 280V when separated by supplementary insulation, and ≥2.0 mm when separated by reinforced insulation	See Table 28.2	N/A
	Distance through reinforced insulation between windings and accessible metal parts was ≥1.0 mm for working voltages ≤ 280V	See Table 28.2	N/A
	Requirement waived as insulation applied was in thin sheet form, other than mica or similar, and for supplementary insulation consisting of at least two layers, one layer having withstood electrical strength test for supplementary insulation		N/A
	Requirement waived as insulation applied was in thin sheet form, other than mica or similar, and for reinforced insulation consisting of at least three layers, two layers having withstood electrical strength test for reinforced insulation		N/A
	Requirement waived as max. temperature rise determined during test of Cl. 12 did not exceed values in 12.5 for inaccessible supplementary or reinforced insulation		N/A
	Requirement waived as inaccessible reinforced or supplementary insulation, after conditioning for 168h at 50 K above max. temperature rise determined per Cl. 12, withstood test of Annex D at the oven temperature and room temperature (°C) ..... :	See Table D.2	N/A
	For optocouplers, 168 h of conditioning at 50 K above the max. temperature rise measured on optocouplers during tests of Clauses 12 and 18, while operating under most difficult conditions		N/A
ANNEX B	**MOTORS NOT ISOLATED FROM THE SUPPLY MAINS AND HAVING BASIC INSULATION NOT DESIGNED FOR THE RATED VOLTAGE OF THE TOOL**		N/A
B.1.1	Motors with working voltage ≤ 42 V		N/A
B.9.2	Metal parts of motor considered bare live parts		N/A
B.12.4	Temperature rise of body of motor determined instead of the temperature rise of the windings		N/A
B.12.5	Temperature rise of the body of the motor in contact with insulating materials did not exceed values in Table 1 for the relevant insulting material	See Table 12.1	N/A
B.18. 201	Tool operated at rated voltage with the terminals of motor and its capacitors short circuited		N/A

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Clause	Requirement + Test	Result - Remark	Verdict
	Tool operated at rated voltage with the supply to the motor open circuited		N/A
	Tool operated at rated voltage with shunt resistor open circuited during operation of motor		N/A
B.21.101	For class I tools with a motor supplied by a rectifier circuit, dc circuit insulated from accessible parts of the tool by double or reinforced insulation		N/A
ANNEX C	**LEAKAGE CURRENT**		N/A
C.2	Leakage current measurement of non-operating tool		N/A
C.3	Leakage current measurement of operating tool		N/A
ANNEX D	**ELECTRIC STRENGTH**		N/A
D.1	Any protective impedance were disconnected		N/A
	The tools were not connected to the supply		N/A
	Electric strength is checked by the tests of D.2		N/A
	For tools with heating elements, test voltages of IEC 60335-1:2010 apply to the heating elements only		N/A
	Insulation between live parts of motor in accordance with Annex B and its other metal parts not subjected to this test		N/A
	Tool in accordance with Annex L, tool is directly connected to the mains or to a non-isolated source		N/A
	Electronic devices bypassed to enable the test to be conducted		N/A
D.2	Test duration 1 min		N/A
	Voltage of substantially sinusoidal waveform, frequency 50 Hz or 60 Hz		N/A
	Electric strength test, voltages applied ..... :		N/A
	To distinguish between capacitor reactance current and unacceptable performance, d.c. potential 1,414 times the that for a.c. was used ..... :		N/A
	No flashover or breakdown occurred during the test		N/A
ANNEX H	**LOW-POWER CIRCUITS**		N/A
	Any points closest to the supply at which the maximum power delivered to the variable resistor does not exceed 15 W at the end of 5 s identified as called a low power points ..... :		N/A
ANNEX K	**BATTERY TOOLS AND BATTERY PACKS**		N/A
K.1	Rated voltage for tools and battery packs ≤75 V d.c.	N/A	P
K.5.7	Tests to be done at rated voltage were done with a fully charged battery		N/A

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Clause	Requirement + Test	Result - Remark	Verdict
K.5.201	Peak voltage of any superimposed ripple exceeding 10 % of the average value was included		N/A
K.5.202	Measurements of lithium-ion cell voltages were made using a filter as specified		N/A
K.5.203	Test area protected against fire and explosion, and well ventilated		N/A
K.5.204	Discharging and charging as specified		N/A
K.5.205	Thermocouples for lithium-ion cell temperature measurement located as specified		N/A
K.5.206	Currents measured during battery charging are average currents		N/A
K.5.207	Fully charged batteries used, after resting for ≥ 2 h but ≤ 6 h at an ambient temperature of (20 ± 5) °C		N/A
K.5.208	Battery consisting of a single cell not subject to special preparations of a cell in a series configuration		N/A
K.5.209	For series arrangement of parallel clusters of cells, the cluster is treated as single cell for specified tests		N/A
K.5.210	End-of-discharge voltages for common cell chemistries observed ..... :		N/A
K.8.3	Battery tools and detachable or separable battery packs marked with additional information		N/A
	- Business name and address of the manufacturer and, where applicable, its authorised representative ..... :		—
	- Designation of series or type ..... :		—
	Battery tools also marked with additional information		N/A
	- Year of manufacture and a date code identifying at least the month of manufacture ..... :		—
	- Designation of the tool ..... :		—
	- identification for parts shipped separately for assembly by the end user		—
	Detachable or separable battery packs marked with additional information		N/A
	- capacity in Ah or mAh .....:		—
	- type of battery.....:		—
	No misunderstanding by additional markings		N/A
K.8.4	Markings specified in K.8.1, 8.2 and K.8.3 not on a **detachable part** of the tool		N/A
	Markings specified in 8.2 clearly discernible from the outside of the tool		N/A
	Markings specified in K.8.3 visible with any separable battery pack or detachable battery pack removed		N/A
	Other markings on the tool visible after removal of a cover		N/A

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Clause	Requirement + Test	Result - Remark	Verdict
K.8.14.1.1	5) Battery tool use and care		N/A
	6) Service		N/A
K.8.14.2	e) Instructions for battery tools		N/A
K.9.1	Construction and enclosure provide adequate protection against electric shock		N/A
K.9.3	No two conductive, simultaneously accessible parts where the voltage between them is hazardous		N/A
	Conductive, simultaneously accessible parts provided with protective impedance		N/A
	Short circuit current between two simultaneously accessible parts (mA) ..... :		N/A
	Capacitance between two simultaneously accessible parts (μF) ..... :		N/A
K.9.5	Electric strength test of D.2 with 750 V applied to insulating material protecting from electric shock	See Table D.2	N/A
K.12.1	Tool operated at no-load until maximum temperature reached or battery discharged ..... :		N/A
	No operation of protective devices during heating test		N/A
	Temperature rises met values in Table 2		N/A
K.12.201	Charging of lithium-ion battery under normal conditions did not exceed specified operating region for charging of the cell		N/A
	Charging procedure as specified		N/A
	Voltage, temperature and charging current monitored for all individual cells		N/A
	Test repeated with imbalanced battery		N/A
K.13.1	Thermoplastic materials of relevant enclosure parts sufficiently resistant to heat		N/A
	Ball-pressure test of IEC 60695-10-2:2003	See Table 13.1	N/A
K.13.2	Glow-wire test applicable only to external enclosure enclosing the current-carrying parts		N/A
	Non-metallic parts in of detachable or separable battery pack supporting connections that carry ≥0,2 A during charging and those within a distance of 3 mm, subjected to the glow-wire test at 850 °C	See Table 13.2	N/A
K.13.2.2101	Polymeric battery enclosure material around current-carrying parts at least classified V according to IEC 60695-11-10:2013, unless …		N/A
	… battery pack was tested to K.18.1 a).		N/A
K.18.1	Risk of fire or electric shock as a result of abnormal operation obviated as far as is practical		N/A

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Clause	Requirement + Test	Result - Remark	Verdict
	No charring or burning of gauze or tissue paper resulted when battery tool and battery pack were subjected to any abnormal operations, tests a) to f)	See Table K.18.1	N/A
	No explosion during or after the test		N/A
	Adequate protection against electric shock		N/A
	Component(s) or conductors(s) that interrupt or limit the discharge current that operated operate during the above tests a) to f) ..... :	See Table K.18.1	N/A
	Test repeated two more times for devices relied upon to pass the test; devices opened the circuit in the same manner		N/A
	Test repeated with the open-circuited device bridged for devices not relied upon to pass the test		N/A
	Protective electronic circuits whose function is relied on to pass a test regarded as providing a SCF and comply with 18.8 with a PL = a	See Table 18.8	N/A
K.18.8	Li-ion charging systems are covered by K.18.201		N/A
K.18.201	Risk of fire and explosion as a result of abnormal operation during charging of a lithium-ion battery is obviated as far as is practical		N/A
	No charring or burning of gauze or tissue paper, no explosion resulted when battery tool and battery pack were subjected to any abnormal conditions a) to d)	See Table K.18.201	N/A
	The cells did not exceed the upper limit charging voltage by more than 150 mV unless…		N/A
	…charging system permanently was disabled from recharging the battery		N/A
	No evident damage to the cell vent to impair compliance with Subclause K.21.202.		N/A
K.18.202	No risk of fire or explosion when main discharge connections of a series configured, integral Li-ion battery, detachable or separable Li-ion battery pack were shorted under extreme imbalance		N/A
	All cells fully charged, one cell fully discharged		N/A
	Main discharge connections of the battery were shorted, resistance ≤10 mΩ		N/A
	No explosion during or after the test		N/A
	No charring or burning of the gauze or tissue paper		N/A
	Component(s) or conductors(s) that interrupt or limit the discharge current that operated operate during the above tests ..... :		N/A
	Test repeated two more times for devices relied upon to pass the test; devices opened the circuit in the same manner		N/A
	Test repeated with the open-circuited device bridged		N/A

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Clause	Requirement + Test	Result - Remark	Verdict
	Protective electronic circuits whose function is relied on to pass a test regarded as providing a SCF and comply with 18.8 with a PL = a	See Table 18.8	N/A
K.18.203	No risk of fire or explosion during abusive overcharging of batteries comprised of cells other than the Li-ion type		N/A
	Battery was charged during 1,25 h at a rate of 10 times the C5 rate for the battery		N/A
	No explosion during or after the test		N/A
	No charring or burning of the gauze or tissue paper		N/A
K.19.6	Marking with rated no-load speed required, measured no-load speed of the spindle did not exceed 110 % of the rated no-load speed		N/A
	No-load speed measured after - operated for 5 min at no-load - replacing the battery with a fully charged battery - operating for 1 min at no-load		N/A
K.19.201	Not possible to install a detachable or separable battery pack in reverse polarity		N/A
K.19.202	Li-ion battery enclosure designed to safely release gases generated as a result of venting		N/A
	Total area of the openings in the enclosure allowing gases to pass without obstruction is ≥ 20 mm²; or…		N/A
	… pressure drop within enclosure was tested , no rupture occurred		N/A
K.20.1	Battery tools and battery packs have adequate mechanical strength and withstand tests of 20.2 and K.20.3.1 or K.20.3.2 and		N/A
	- did not catch fire or explode		N/A
	- met requirements of clauses K.9, K.19 and either K.18.1 (f) or K.28.1 after tests of 20.2 and 28.1		N/A
	Li-ion battery tools and battery packs, after the test of K.20.3.1 or K.20.3.2, - did not have an open circuit voltage below 90 % of the voltage measured immediately prior to the test		N/A
	- demonstrated normal discharging and recharging after the test		N/A
	- showed no damage to the cell vent impairing compliance with K.21.202		N/A
K.20.3.1	Adequate mechanical strength after drop tests on a concrete surface from a height of 1 m		N/A
	Test repeated with the battery pack removed from the tool		N/A
	Test repeated on the battery pack by itself		N/A
	The test was repeated with each attachment or combination of attachments		N/A

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Clause	Requirement + Test	Result - Remark	Verdict
K.20.3.2	Impact test with 50 mm, 0,55 kg smooth steel sphere for battery-operated transportable tools ..... :		N/A
	travel of the sphere was 1,3 m..... :		N/A
	Damage (except to a guard) accepted, tool became incapable of normal operation		N/A
	Test repeated separately on detachable or separable battery packs with a mass ≥3 kg		N/A
	Additional drop test on detachable or separable battery packs with a mass <3 kg		N/A
K.21.17.1.2	The number of cycles is 6 000		N/A
K.21.201	Tool will not accept general purpose batteries as an energy source for their primary function		N/A
K.21.202	Venting of lithium-ion cells, if relied on for safety, not adversely obstructed		N/A
K.21.203	Unsuitable connector types not used for user accessible interfaces between elements of a Li-ion battery system		N/A
K.23.1.10	Power switches have adequate breaking capacity and present no electrical or mechanical failure		N/A
	50 cycles of making and breaking the locked output mechanism current		N/A
K.23.1.201	Power switches withstood, without excessive wear or other harmful effect, the mechanical, electrical, and thermal stresses occurring in normal use		N/A
	6000 cycles of operation making and breaking the no-load of the tool at a fully charged battery		N/A
K.23.201	Battery cells comply with IEC 62133		N/A
K.23.202	Rechargeable battery cells not of lithium-metal type		N/A
K.24.201	External flexible cable or cord of battery tools with separable battery packs have anchorages such that the conductors are relieved from strain, including twisting, where they are connected within the tool, and protected from abrasion		N/A
K.28.1	Creepage distances and clearances not less than the values in millimetres shown in Table K.1 ..... :	See Table 28.1	N/A
	Smaller clearance and creepage distances for parts of different polarity accepted, shorting of the two parts did not result in the tool starting		N/A
	For parts with a hazardous voltage between them, the sum total of the measured distances between each of these parts and their nearest accessible surface is not less than 1,5 mm clearance and 2,0 mm creepage (Fig. K.1) ..... :		N/A
	Creepage distances and clearances measured as indicated in Annex A		N/A

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Clause	Requirement + Test	Result - Remark	Verdict
	Distances through slots or openings in external parts of insulating material measured to metal foil in contact with the accessible surface		N/A
	Foil pushed into corners and the like by means of test probe B of IEC 61032:1997, except not pressed into openings		N/A
	The sum total of distances measured between parts operating at hazardous voltage and accessible surfaces determined by measuring the distance from each part to the accessible surface		N/A
	Distances added together to determine the sum total (see Figure K.1)		N/A
	One of the distances was 1,0 mm or greater (see Annex A, cases 1 to 10)		N/A
	Force applied by means of test probe B of IEC 61032:1997 at the following values:		N/A
	– 2 N for bare conductors		N/A
	– 30 N for enclosures		N/A
	Means provided for securing the tool to a support considered to be accessible		N/A

ANNEX L	**BATTERY TOOLS AND BATTERY PACKS PROVIDED WITH MAINS CONNECTION OR NON-ISOLATED SOURCES**		N/A
L.1	Rated voltage for battery pack ≤250 V a.c. (single phase) or d.c. mains source and ≤75 V d.c. battery source		N/A
	Rated voltage for battery pack ≤75 V d.c.		N/A
L.5.7	Tests to be done at rated voltage were done with a fully charged battery		N/A
L.5.201	Peak voltage of any superimposed ripple exceeding 10 % of the average value was included		N/A
L.5.202	Measurements of lithium-ion cell voltages were made using a filter as specified		N/A
L.5.203	Test area protected against fire and explosion, and well ventilated		N/A
L.5.204	Discharging and charging as specified		N/A
L.5.205	Thermocouples for lithium-ion cell temperature measurement located as specified		N/A
L.5.206	Currents measured during battery charging are average currents		N/A
L.5.207	Fully charged batteries used, after resting for ≥ 2 h but ≤ 6 h at an ambient temperature of (20 ± 5) °C		N/A
L.5.208	Battery consisting of a single cell not subject to special preparations of a cell in a series configuration		N/A

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Clause	Requirement + Test	Result - Remark	Verdict
L.5.209	For series arrangement of parallel clusters of cells, the cluster is treated as single cell for specified tests		N/A
L.5.210	End-of-discharge voltages for common cell chemistries observed ..... :		N/A
L.8.1	Non-isolated sources that can supply a tool, or tool that can be supplied directly from the mains, marked with as required by the standard:		N/A
	Rated voltage(s) or voltage range(s), (V) ..... :		—
	Symbol for nature of supply or frequency (Hz)..... :		—
	Rated input, (W) or rated current (A) ..... :		—
	Symbol for class II ..... :		—
L.8.3	Tools and detachable or separable battery packs marked with additional information		N/A
	- Business name and address of the manufacturer and, where applicable, its authorised representative ..... :		—
	- Designation of series or type ..... :		—
	Tools also marked with additional information		N/A
	- Year of manufacture and a date code identifying at least the month of manufacture ..... :		—
	- Designation of the tool ..... :		—
	- identification for parts shipped separately for assembly by the end user		—
	Detachable or separable battery packs marked with additional information		N/A
	- capacity in Ah or mAh .....:		—
	- type of battery.....:		—
	No misunderstanding by additional markings		N/A
L.8.4	Markings of L.8.1, 8.2 and L.8.3 not on a detachable part of the tool		N/A
	Markings of 8.2 clearly discernible from outside the tool		N/A
	Markings of L.8.3 visible with any separable or detachable battery pack removed		N/A
	Other markings may be visible after removing cover		N/A
	Indications for switches and controls placed on or in vicinity of components		N/A
	Not placed on parts which can be repositioned		N/A
	Not positioned such that making the marking is misleading		N/A
L.8.14.1.1	5) Battery tool use and care		N/A
	6) Service		N/A

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Clause	Requirement + Test	Result - Remark	Verdict
L.8.14.2	e) Instructions for battery tools		N/A
L.9	Construction and enclosure provide adequate protection against electric shock		N/A
	Tools connected to the mains or supplied by a non-isolated source.		N/A
	Tool also evaluated with the battery pack removed when removal without the use of a tool was possible		N/A
L.9.201	There are no two conductive simultaneously accessible parts where the voltage between them is hazardous, except when provided with protective impedance		N/A
	Short circuit current between two simultaneously accessible parts (mA) ..... :		N/A
	Capacitance between two simultaneously accessible parts (μF) ..... :		N/A
L.10	Applied only when tool is directly connected to mains, or to a non-isolated source		N/A
L.11	Applied only when tool is directly connected to mains, or to a non-isolated source		N/A
	Test on tool capable of charging the battery while performing its function conducted while charging a discharged battery pack		N/A
L.12	Applied only when tool directly connected to mains, or to a non-isolated source		N/A
	Test on tool capable of charging the battery while performing its function conducted while charging a previously discharged battery pack with the charger connected		N/A
	Tool operated at no-load until maximum temperature reached or battery discharged ..... :		N/A
	Test repeated, allowing the battery pack to charge while the tool was not operating		N/A
L.12.201	Charging of lithium-ion battery under normal conditions did not exceed specified operating region for charging of the cell		N/A
	Charging procedure as specified		N/A
	Voltage, temperature and charging current monitored for all individual cells		N/A
	Test repeated with imbalanced battery		N/A
L.13.1	Applied only when tool directly connected to mains, or to a non-isolated source	See Table 13.1	N/A
	Tool capable of charging the battery while performing its function also evaluated with charger connected to the mains		N/A
	Tool also evaluated with battery power alone when more unfavourable temperatures may result		N/A

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Clause	Requirement + Test	Result - Remark	Verdict
L.13.2	Non-metallic parts in of detachable or separable battery pack supporting connections that carry ≥0,2 A during charging and those within a distance of 3 mm, subjected to the glow-wire test at 850 °C	See Table 13.2	N/A
L.14	Applied only when tool directly connected to mains, or to a non-isolated source		N/A
L.16	Applied only when tool directly connected to mains, or to a non-isolated source		N/A
L.17	Applied only when tool directly connected to mains, or to a non-isolated source		N/A
	Tools not capable of continuous operation operated under battery power for the duration of the test, except evaluated for electric strength with their charger connected		N/A
L.18	Applied only when tool directly connected to mains, or to a non-isolated source, except L.18.8 and L.18.201 to L.18.204,		N/A
L.18.8	Applied only to charging systems other than Li-ion		N/A
L.18.201	Risk of fire or electric shock as a result of abnormal operation obviated as far as is practical		N/A
	No charring or burning of gauze or tissue paper resulted when battery tool and battery pack were subjected to any abnormal operations, tests a) to f)	See Table L.18.201	N/A
	No explosion during or after the test		N/A
	Adequate protection against electric shock		N/A
	Component(s) or conductors(s) that interrupt or limit the discharge current that operated operate during the above tests a) to f) ..... :	See Table L.18.201	N/A
	Test repeated two more times for devices relied upon to pass the test; devices opened the circuit in the same manner		N/A
	Test repeated with the open-circuited device bridged		N/A
	Protective electronic circuits whose function is relied on to pass a test regarded as providing a SCF and comply with 18.8 with a PL = a	See Table 18.8	N/A
L.18.202	Risk of fire and explosion as a result of abnormal operation during charging of a lithium-ion battery is obviated as far as is practical		N/A
	No charring or burning of gauze or tissue paper, no explosion resulted when battery tool and battery pack were subjected to any abnormal conditions a) to d)	See Table L.18.202	N/A
	The cells did not exceed the upper limit charging voltage by more than 150 mV unless…		N/A
	…charging system permanently was disabled from recharging the battery		N/A

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Clause	Requirement + Test	Result - Remark	Verdict
	No evident damage to the cell vent to impair compliance with Subclause K.21.202.		N/A
L.18.203	No risk of fire or explosion when main discharge connections of a series configured, integral Li-ion battery, detachable or separable Li-ion battery pack were shorted under extreme imbalance		N/A
	All cells fully charged, one cell fully discharged		N/A
	Main discharge connections of the battery were shorted, resistance ≤10 mΩ		N/A
	No explosion during or after the test		N/A
	No charring or burning of the gauze or tissue paper		N/A
	Component(s) or conductors(s) that interrupt or limit the discharge current that operated operate during the above tests ..... :		N/A
	Test repeated two more times for devices relied upon to pass the test; devices opened the circuit in the same manner		N/A
	Test repeated with the open-circuited device bridged		N/A
	Protective electronic circuits whose function is relied on to pass a test regarded as providing a SCF and comply with 18.8 with a PL = a	See Table 18.8	N/A
L.18.204	No risk of fire or explosion during abusive overcharging of batteries comprised of cells other than the Li-ion type		N/A
	Battery was charged during 1,25 h at a rate of 10 times the C5 rate for the battery		N/A
	No explosion during or after the test		N/A
	No charring or burning of the gauze or tissue paper		N/A
L.19.201	Not possible to connect a battery pack in reverse polarity		N/A
L.19.202	Li-ion battery enclosure designed to safely release gases generated as a result of venting		N/A
	Total area of the openings in the enclosure allowing gases to pass without obstruction is ≥ 20 mm²; or…		N/A
	… pressure drop within enclosure was tested , no rupture occurred		N/A
L.20	Applied only when tool directly connected to mains, or to a non-isolated source, except L.20.201 and L.20.202		N/A
L.20.201	Battery tools with its battery pack attached have adequate mechanical strength and withstand tests of L.9, L.19, L.28.1 and either L.18.201 f) or L.28.201, and		N/A
	- did not catch fire or explode		N/A
	- demonstrated normal discharging and recharging after the test		N/A

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Clause	Requirement + Test	Result - Remark	Verdict
	- showed no damage to the cell vent impairing compliance with L.21.202		N/A
L.20.202	For hand-held battery tools, L.20.202.1 applies; for transportable battery tools, L.20.202.2 applies		N/A
L.20.202.1	Adequate mechanical strength after drop tests on a concrete surface from a height of 1 m		N/A
	Test repeated with the battery pack removed from the tool		N/A
	Test repeated on the battery pack by itself		N/A
	The test was repeated with each attachment or combination of attachments		N/A
L.20.202.2	Impact test with 50 mm, 0,55 kg smooth steel sphere for battery-operated transportable tools ..... :		N/A
	travel of the sphere was 1,3 m..... :		N/A
	Damage (except to a guard) accepted, tool became incapable of normal operation		N/A
	Test repeated separately on detachable or separable battery packs with a mass ≥3 kg		N/A
	Additional drop test on detachable or separable battery packs with a mass <3 kg		N/A
L.21	Applied only when tool directly connected to mains, or to a non-isolated source, except L.21.201 and L.21.202		N/A
L.21.201	Tool will not accept general purpose batteries as an energy source for their primary function		N/A
L.21.202	Venting of lithium-ion cells, if relied on for safety, not adversely obstructed		N/A
L.21.203	Unsuitable connector types not used for user accessible interfaces between elements of a Li-ion battery system		N/A
L.22	Applied only when tool directly connected to mains, or to a non-isolated source		N/A
L.23	Components		
L.23.1.10	Applied only to power switches of tools capable of performing their intended operation when connected to the mains or to a non-isolated source		N/A
L.23.1.10.201	Switches controlling the primary operating means of the tool, except as indicated in L.23.1.10, have adequate breaking capacity and presented no electrical or mechanical failure		N/A
L.23.1.10.202	Power switches withstood, without excessive wear or other harmful effect, the mechanical, electrical, and thermal stresses occurring in normal use		N/A
	6000 cycles of operation making and breaking the no-load of the tool at a fully charged battery		N/A
L.23.201	Battery cells comply with IEC 62133		N/A

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Clause	Requirement + Test	Result - Remark	Verdict
L.23.202	Rechargeable battery cells not of lithium-metal type		N/A
L.24.1	Also applied to the flexible cord between a non-isolated power source and the tool		N/A
L.24.3	Also applied to the flexible cord between a non-isolated power source and the tool		N/A
L.24.4	This subclause applied, except flexible cord provided between a non-isolated power source and the tool not provided with a plug that can be connected directly to the mains		N/A
L.24.5	Not applied to flexible cord provided between a non-isolated power source and the tool		N/A
L.24.20	Requirements of this Subclause applied, except the flexible cord between a non-isolated power source and the tool not provided with an appliance inlet that can allow direct connection to mains		N/A
L.24.201	External flexible cable and cord have anchorages such that the conductors are relieved from strain, including twisting, where they are connected within the tool, and protected from abrasion		N/A
L.25	Not applied to interconnecting cords		N/A
L.26	Applied to the tool directly connected to the mains or to a non-isolated source		N/A
L.28.1	Applied when tool is directly connected to the mains or to a non-isolated source		N/A
	Battery packs connected to the tool during the evaluation		N/A
	Tool also evaluated with the battery pack removed when the removal could be accomplished without the use of a tool		N/A
	Creepage distances and clearances of IEC 60335-1: 2010 applied as applicable		N/A
L.28.201	Creepage distances and clearances not less than the values in millimetres shown in Table L.1		N/A
	Smaller clearance and creepage distances for parts of different polarity accepted, shorting of the two parts did not result in the tool starting		N/A
	For parts having a hazardous voltage between them, the sum of the measured distances between each of these parts and their nearest accessible surface is not less than 1.5 mm clearance and 2.0 mm creepage (Fig. L.1)		N/A
	Creepage distances and clearances measured as indicated in Annex A		N/A
	Distances through slots or openings in external parts of insulating material measured to metal foil in contact with the accessible surface		N/A

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Clause	Requirement + Test	Result - Remark	Verdict
	Foil pushed into corners and the like by means of test probe B of IEC 61032:1997, except not pressed into openings		N/A
	The sum total of distances measured between parts operating at hazardous voltage and accessible surfaces determined by measuring the distance from each part to the accessible surface		N/A
	Distances added together to determine the sum total (see Figure L.1)		N/A
	One of the distances was 1,0 mm or greater (see Annex A, cases 1 to 10)		N/A
	Force applied by means of test probe B of IEC 61032:1997 at the following values:		N/A
	– 2 N for bare conductors		N/A
	– 30 N for enclosures		N/A
	Means provided for securing the tool to a support considered to be accessible		N/A

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Clause	Requirement + Test	Result - Remark	Verdict

8.12 A	TABLE: Label Heating Test			N/A
Test Conditions ..... :	Conditioned in oven for 24 h at (120 ± 2) °C Y/N Conditioned in oven for 200 h at: °C Y/N Amount of samples: 3			
Test Specimen	Material type	Good adhesion and no curling of edges	Label resists defacement and removal when scraped	
Rating label				
Supplementary information:				

8.12 B	TABLE: Label immersion tests – Water			P
Test Conditions ..... :	Pre Treating for 24 h at relative humidity of 45 % and temperature: °C Time of labels in water: 48 h Amount of samples: 3			
Test Specimen	Material type	Good adhesion and no curling of edges	Label resists defacement and removal when scraped	
Rating label	--	Yes	N/A	
Supplementary information:				

8.12 C	TABLE: Label immersion tests - Oil (IRM 903)			P
Test Conditions ..... :	Pre Treating for 24 h at relative humidity of 45 % and temperature: °C Time of labels in oil: 48 h Amount of samples: 3			
Test Specimen	Material type	Good adhesion and no curling of edges	Label resists defacement and removal when scraped	
Rating label	--	Yes	N/A	
Supplementary information:				

8.12 D	TABLE: Label Standard atmosphere tests			P
Test Conditions ..... :	Time of labels in controlled atmosphere at relative humidity of 45%: 72h Controlled atmosphere temperature: °C Amount of samples: 3			
Test Specimen	Material type	Good adhesion and no curling of edges	Label resists defacement and removal when scraped	
Rating label	--	Yes	N/A	
Supplementary information:				

UL 62841-1			
Clause	Requirement + Test	Result - Remark	Verdict

9.1	TABLE: Protection against access to live parts			N/A
Measurement between relevant parts and poles of supply source	Rated voltage U (V)	Measured voltage (V)	Measured current (A)	Measured capacitance (μF)
Supplementary information:				

11	TABLE: Input data under no-load conditions				N/A
Input deviation of/at:	**Rated P (W) or I (A)**	**Measured P (W) or I (A)**	**Ratio (%)**	**Required ratio (%)**	**Remark**
Supplementary information:					

UL 62841-1			
Clause	Requirement + Test	Result - Remark	Verdict

12.1A	TABLE: Temperature rise measurements under the conditions of 12.2 to 12.5		N/A
Test voltage (V)..... :			—
Ambient temperature, t_1 (°C) :			—
Ambient temperature, t_2 (°C) :			
Operating time (min, s) ..... :			—
Speed (min^{-1})..... :			—
Input Wattage (W) ..... :			—
Input current (A) ..... :			—
Torque (Nm) ..... :			—

Thermocouple Locations	Temperature rise measured (K)	Temperature rise limit (K)
Supplementary information:		

UL 62841-1			
Clause	Requirement + Test	Result - Remark	Verdict

12.1B	TABLE: Heating test, resistance method		N/A
	Test voltage (V) ..... :		—
	Ambient, t_1 (°C) .....:		—
	Ambient, t_2 (°C) .....:		—

Temperature rise of winding	R_1 (Ω)	R_2 (Ω)	ΔT measured (K)	ΔT Limit (K)	Insulation class
Stator (1)					
Stator (2)					
Rotor (2)					
Supplementary information:					

13.1	TABLE: Ball Pressure Test of Thermoplastics			N/A
Allowed impression diameter (mm) ..... :			2,0	—
Object/ Part No.	Material	Manufacturer/ trademark	Test temperature (°C)	Impression diameter (mm)
Supplementary information:				

13.2	TABLE: Glow Wire Test					N/A
Object/ Part No.	Material	Manufacturer/ trademark	Test temperature (ºC)	Material ignited, Yes/No	Layer under Test Sample ignited, Yes/No	Verdict
Supplementary information:						

UL 62841-1			
Clause	Requirement + Test	Result - Remark	Verdict

16	TABLE: Overload Protection of Transformers and Associated Circuits		N/A
Test voltage :			—
Ambient temperature (°C) :			—
Input current (A) / Input Wattage (W) :			—
Applied short-circuit or overload :			—
Measurement at:		Temperature rise (K)	Allowed Limit (K)
Transformer winding (thermocouple)			
Transformer winding (T_1)R-R			
Transformer winding (T_2)R-R			
Transformer Lamination			
Internal wiring			
Capacitor			
Printed circuit board			
SELV circuits			
Supplementary Information:			

18.6.1	TABLE: Fault Condition Tests			N/A	
	Ambient temperature (°C)..... :				—
	Fuse-link Current (A)				—
Component	Fault Condition	Test Voltage (V)	Test Duration*	Comment/Result Test repeated Yes/No**	

Supplementary Information:
* Tests were continued until
- a protective device operates, or
- until steady conditions are established or
- an open circuit occurs.

** Test was repeated on a second sample due to an intentionally weak part permanently open-circuited to terminate the test.

18.8.1A	TABLE: Performance levels of Safety Critical Functions		N/A
Type and purpose of SCF	Min. PL determined based on:[1,2]	Min. PL	Actual PL

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Clause	Requirement + Test	Result - Remark	Verdict

Supplementary Information:
[1] Relevant part of IEC 62841-2, IEC 62841-3 or IEC 62841-4 or; if no such part existent, ISO 13849-1 using Annex E as a guide

[2] For safety critical functions not listed in Table 4 of IEC 62841-1 and provided by electronic circuits, PL values were determined using the methods of ISO 13849-1.

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Clause	Requirement + Test	Result - Remark	Verdict

18.8.1B	**TABLE: Software in Safety Critical Functions**		—
H.11.12.3 from IEC 60730-1:2010			
H.11.12.3	Measures to avoid errors		—
H.11.12.3.1	For controls with software Class B or C the V-model for the software life cycle was applied		N/A
	Measures used for software class C are inherently acceptable for software class B		N/A
	Other methods applied if they incorporate disciplined and structured processes including design and test phases ..... :		N/A
H.11.12.3.2	Specification		—
H.11.12.3.2.1	Software safety requirements		—
H.11.12.3.2.1.1	The specification of the software safety requirements includes:		—
	• A description of each safety related function to be implemented, including its response time(s): ○ functions related to the application including their related software classes ○ functions related to the detection, annunciation and management of software or hardware faults		N/A
	• A description of interfaces between software and hardware		N/A
	• A description of interfaces between any safety and non-safety related functions		N/A
H.11.12.3.2.2	Software architecture		—
H.11.12.3.2.2.1	The description of software architecture shall include the following aspects:		—
	• Techniques and measures to control software faults/errors (refer to H.11.12.2)		N/A
	• Interactions between hardware and software		N/A
	• Partitioning into modules and their allocation to the specified safety functions		N/A
	• Hierarchy and call structure of the modules (control flow)		N/A
	• Interrupt handling		N/A
	• Data flow and restrictions on data access		N/A
	• Architecture and storage of data		N/A
	• Time based dependencies of sequences and data		N/A

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Clause	Requirement + Test	Result - Remark	Verdict
H.11.12.3.2.2.2	The architecture specification was verified against the specification of the software safety requirements by static analysis. Acceptable methods are:		—
	• Control flow analysis		N/A
	• Data flow analysis		N/A
	• Walk-throughs / design reviews		N/A
H.11.12.3.2.3.1	Based on the architecture design, software is suitably refined into modules. Software module design and coding are implemented in a way that is traceable to the software architecture and requirements		N/A
H.11.12.3.2.3.2	Software code is structured		N/A
H.11.12.3.2.3.3	Coded software is verified against the module specification, and the module specification is verified against the architecture specification by static analysis		N/A
H.11.12.3.2.4	Design and coding standards		—
	Program design and coding standards is consequently used during software design and maintenance		N/A
	Coding standards specify programming practice, proscribe unsafe language features, and specify procedures for source code documentation as well as for data naming conventions		N/A
H.11.12.3.3	Testing		—
H.11.12.3.3.1	Module design (software system design, software module design and coding)		—
H.11.12.3.3.1.1	A test concept with suitable test cases is defined based on the module design specification.		N/A
H.11.12.3.3.1.2	Each software module is tested as specified within the test concept		N/A
H.11.12.3.3.1.3	Test cases, test data and test results are documented		N/A
H.11.12.3.3.1.4	Code verification of a software module by static means includes such techniques as software inspections, walk-throughs, static analysis and formal proof		N/A
	Code verification of a software module by dynamic means includes functional testing, white-box testing and statistical testing		N/A
H.11.12.3.3.2	Software integration testing		N/A
H.11.12.3.3.2.1	A test concept with suitable test cases is defined based on the architecture design specification		N/A

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Clause	Requirement + Test	Result - Remark	Verdict
H.11.12.3.3.2.2	The software is tested as specified within the test concept		N/A
H.11.12.3.3.2.3	Test cases, test data and test results are documented		N/A
H.11.12.3.3.3	Software validation		—
H.11.12.3.3.3.1	A validation concept with suitable test cases is defined based on the software safety requirements specification		N/A
H.11.12.3.3.3.2	The software is validated with reference to the requirements of the software safety requirements specification as specified within the validation concept.		N/A
	The software is exercised by simulation or stimulation of:		N/A
	• input signals present during normal operation		N/A
	• anticipated occurrences		N/A
	• undesired conditions requiring system action		N/A
H.11.12.3.3.3.4	Test cases, test data and test results are documented		N/A
H.11.12.3.4	Other Items		—
H.11.12.3.4.1	Tools, programming languages are assumed to be suitable if they comply with "increased confidence from use" according to IEC 61508-7, C.4.4	Only applicable for SCF with PL ≥ c	N/A
H.11.12.3.4.2	Management of software versions: All versions are uniquely identified for traceability		N/A
H.11.12.3.4.3	Software modification		—
H.11.12.3.4.3.1	Software modifications are based on a modification request which details the following:		—
	• the hazards which may be affected		N/A
	• the proposed change		N/A
	• the reasons for change		N/A
H.11.12.3.4.3.2	An analysis is carried out to determine the impact of the proposed modification on functional safety.		N/A
H.11.12.3.4.3.3	A detailed specification for the modification is generated including the necessary activities for verification and validation, such as a definition of suitable test cases		N/A
H.11.12.3.4.3.4	The modification are carried out as planned		N/A
H.11.12.3.4.3.5	The assessment of the modification is carried out based on the specified verification and validation activities. This may include:		N/A

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Clause	Requirement + Test	Result - Remark	Verdict
	• a reverification of changed software modules		N/A
	• a reverification of affected software modules		N/A
	• a revalidation of the complete system		N/A
H.11.12.3.4.3.6	All details of modification activities are documented		N/A
H.11.12.3.5	For class C control functions: One of the combinations (a–p) of analytical measures given in the columns of table H.9 is used during hardware development .....:	Measures to avoid errors for class C not required	N/A

23.1	TABLE: Critical components information				N/A
Object / part No.	**Manufacturer/ trademark**	**Type / model**	**Technical data**	**Standard**	**Mark(s) of conformity[1)]**
[1)] Provided evidence ensures the agreed level of compliance. See OD-CB2039.					

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Clause	Requirement + Test	Result - Remark	Verdict

24.1	TABLE: Length of supply cord			N/A
Manufacturer of Cable	Cable type	Nominal cross-section used (mm^2)	Length of supply cord measured (m)	
Supplementary information:				

24.5	TABLE: Nominal cross-section area of supply cord				N/A
Rated current (A)[1]:		Current measured during clause 12ff. (A):		Nominal cross-section required per table 6:	
Manufacturer of Cable	**Cable Type**	**Nominal cross-section used**			
Supplementary information: [1] Current measured during test of clause 12.1, if no current rating marked.					

24.11	TABLE: Flexing and lifting				N/A
Weight of tool (kg):		> 10 000 flexings, sample turned about 90° (Yes/No):			
Weight attached to cable or cord (kg):		Cord guard slipped out after 10 completed 1 s lifts (Yes/No):			
Manufacturer of Cable	Cable type	No. of strands (total)	No. of strands (broken)	Deviation in %	
Supplementary information:					

24.12	TABLE: Cord guard						N/A
Cable manufacturer	Cable type	Overall Ø of cord (mm)	Cord guard length min. (mm)	Cord guard length meas. (mm)	Mass attached (g)	Radius of curvature (mm)	
Supplementary information:							

24.13	TABLE: Cord anchorage					N/A
Manufacturer of Cable	Cable type	Cord is pulled 25 times at (N)	Cord is twisted for 1 min at (Nm)	Cord damaged	Longitudinal displacement	Conductors movement
Supplementary information: -						

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Clause	Requirement + Test	Result - Remark	Verdict

27.1	TABLE: Torque Test for screws and nuts				N/A
Threaded part identification	**Thread diameter (mm)**	**Column number (I, II, or III)**	**Applied torque (Nm)**	**Number of cycles (5 or 10)**	
Fixed enclosure					
Supplementary information:					

28.1	TABLE: Clearance And Creepage Distance Measurements					N/A
clearance cl and creepage distance cr at/of:	**Up (V)**	**U r.m.s. (V)**	**Required cl (mm)**	**cl (mm)**	**required cr (mm)**	**cr (mm)**
Supplementary information:						

28.2	TABLE: Distance Through Insulation Measurements				N/A
Distance through insulation dti at/of:	**U r.m.s. (V)**	**Test voltage (V)**	**Required dti (mm)**	**dti (mm)**	
Supplementary information:					

C.2A	TABLE: Leakage Current of the non-operating tool as per clause 14.1				N/A
Points of application	Test voltage (rated V)	Freq. (Hz)	Selector Switch Position	Allowed leakage current (mA)	Measured leakage (mA)
Supplementary Information:					

C.2B	TABLE: Leakage Current of the non-operating tool as per clause 14.4				N/A
Points of application	Test voltage (rated V)	Freq. (Hz)	Selector Switch Position	Allowed leakage current (mA)	Measured leakage (mA)
Supplementary Information:					

UL 62841-1			
Clause	Requirement + Test	Result - Remark	Verdict

C.3A	TABLE: Leakage Current of the operating tool as per clause 12.1				N/A
Points of application	Test voltage (1.06 X rated V)	Freq. (Hz)	Selector Switch Position (ON /OFF[1])	Allowed leakage current (mA)	Measured leakage (mA)
Supplementary Information:					

C.3B	TABLE: Leakage Current of the operating tool as per clause 14.3				N/A
Points of application	Test voltage (rated V)	Freq. (Hz)	Selector Switch Position (ON /OFF[1])	Allowed leakage current (mA)	Measured leakage (mA)
Supplementary Information:					

C.3C	TABLE: Leakage Current of the operating tool as per clause 18.5.1				N/A
Points of application	Test voltage (rated V)	Freq. (Hz)	Selector Switch Position (ON /OFF[1])	Allowed leakage current (mA)	Measured leakage (mA)
Supplementary Information:					

C.3D	TABLE: Leakage Current of the operating tool as per clause 18.5.4				N/A
Points of application	Test voltage (rated V)	Freq. (Hz)	Selector Switch Position (ON /OFF[1])	Allowed leakage current (mA)	Measured leakage (mA)
Supplementary Information:					

D.2	TABLE: Dielectric Strength		N/A
Test voltage applied between:	**Test during or after clause**	**Test potential applied (V)**	**Breakdown / flashover (Yes/No)**
- windings and metal core of the motor field over **basic insulation**			
- commutator and metal core of the motor armature over **basic insulation**			
- metal core and motor armature spindle of the motor armature over **supplementary insulation**			

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Clause	Requirement + Test	Result - Remark	Verdict

- commutator and motor armature spindle over **reinforced insulation**			
- between live parts and other metal parts over **basic insulation**			
- between inaccessible metal parts and accessible parts over **supplementary insulation**			
- between live parts and accessible parts over **reinforced insulation**			
- accessible metal parts in class I tools and the supply cord wrapped with metal foil			
- accessible metal parts in class II tools and the supply cord wrapped with metal foil			
- between live parts and other metal parts over **basic insulation**			
- between inaccessible metal parts and accessible parts over **supplementary insulation**			
- between live parts and accessible parts over **reinforced insulation**			
- live parts and accessible parts over **basic insulation**			
- live parts and accessible parts over **reinforced insulation**			
- live parts and accessible parts over **basic insulation**			
- live parts and accessible parts over **reinforced insulation**			
- between live parts and other metal parts over **basic insulation**			
- between inaccessible metal parts and accessible parts over **supplementary insulation**			
- between live parts and accessible parts over **reinforced insulation**			
- live parts and accessible parts over **basic insulation**			
- live parts and accessible parts over **reinforced insulation**			
- live parts and accessible parts not grounded, if the tool does not operate anymore			
- live parts and accessible parts not grounded, if the tool still operates			
- commutator segments and armature shaft in series motors with class II armature construction			
- live parts and accessible parts not grounded, if any winding is open circuited			
- live parts and accessible parts not grounded, if no windings are open circuited			
- live parts and accessible parts over **basic insulation**			

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Clause	Requirement + Test	Result - Remark	Verdict

- live parts and accessible parts over **reinforced insulation**			
- between the handles and grasping surfaces in contact with foil and the output shaft of the tool			
- between live parts and other metal parts over **basic insulation**			
- between inaccessible metal parts and accessible parts over **supplementary insulation**			
- between live parts and accessible parts over **reinforced insulation**			
- shafts of operating knobs, handles, levers etc. and their insulating covering wrapped in metal foil			
- live parts and accessible parts over **basic insulation**			
- live parts and accessible parts over **reinforced insulation**			
- basic insulation			
- supplementary insulation			
- reinforced insulation			
- over insulation protecting from electric shock			
Supplementary information:			

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Clause	Requirement + Test	Result - Remark	Verdict

K12.1	TABLE: Normal Temperature Test for Battery Tool		N/A
Ambient temperature (°C)			—
Measurement at:		Temperature rise (K)	Allowed Limit (K)
Enclosure, outside, gripping surface			
Enclosure , outside, near motor			
Enclosure outside, gear housing			
Enclosure, inside, near motor			
Enclosure, inside, near heat sink			
Internal wiring			
Switch body			
External, metal gear case			
Battery terminal support			
Battery pack			
Supplementary Information: Status of overload protector at end of test [ ] No change [ ] Opened during the Test [X] N/A			

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Clause	Requirement + Test	Result - Remark	Verdict

K.18.1	TABLE: Battery Tool Abnormal Operation					N/A
		Resistance applied (max. 10 mΩ)	Protective device operated during first test?	Test repeated 2 more times with device in place?	Test repeated 1 more time with device bridged?	Explosion, charring or burning of test materials?
a) Terminals of detachable battery pack with exposed terminals shorted						
b) Motor terminals shorted						
c) Motor rotor locked						
d) Cord between battery tool and separable battery pack shorted						
e) Cord between tool and charger shorted						
f) Any two uninsulated parts of opposite polarity in battery tools shorted						
Supplementary Information:						

K.18.201	TABLE: Lithium-ion charging systems – Abnormal Conditions					N/A
Abnormal conditions		Explosion occurred?	Charring or burning of test materials?	Upper limit charging voltage not exceeded by >150 mV[1)]	Charging system permanently disabled?[2)]	Cell vent damaged?
a) Components in the charging system faulted as in 18.6.1 b) to f)						
b) One cell 50% charged in a fully discharged battery						
c) Charging of a series configured battery with all cells 50% charged, one cell shorted						
d) Short across a component or between adjacent PCB tracks						
Supplementary Information: One of conditions [1)] or [2)] is sufficient to achieve compliance with this subclause.						

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Clause	Requirement + Test	Result - Remark	Verdict

L.18.1	TABLE: Battery Tool Abnormal Operation					N/A
Abnormal conditions	Resistance applied (max. 10 mΩ)	Protective device operated during first test?	Test repeated 2 more times with device in place?	Test repeated 1 more time with device bridged?	Explosion, charring or burning of test materials?	
a) Terminals of detachable battery pack with exposed terminals shorted						
b) Motor terminals shorted						
c) Motor rotor locked						
d) Cord between battery tool and separable battery pack shorted						
e) Cord between tool and charger shorted						
f) Any two uninsulated parts of opposite polarity in battery tools shorted						
Supplementary Information:						

L.18.202	TABLE: Lithium-ion charging systems – Abnormal Conditions					N/A
Abnormal conditions	Explosion occurred?	Charring or burning of test materials?	Upper limit charging voltage not exceeded by >150 mV[1)]	Charging system permanently disabled?[2)]	Cell vent damaged?	
a) Components in the charging system faulted as in 18.6.1 b) to f)						
b) One cell 50% charged in a fully discharged battery						
c) Charging of a series configured battery with all cells 50% charged, one cell shorted						
d) Short across a component or between adjacent PCB tracks						
Supplementary Information: One of conditions [1)] or [2)] is sufficient to achieve compliance with this subclause.						

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Clause	Requirement + Test	Result - Remark	Verdict

ATTACHMENT TO TEST REPORT UL 62841-1

EUROPEAN GROUP DIFFERENCES AND NATIONAL DIFFERENCES

Electric Motor-Operated Hand-Held, Transportable Tools and Lawn and Garden Machinery - Safety - Part 1: General requirements

Differences according to ..... :	UL 62841-1:2015 + AC.2015
Attachment Form No..... :	UL 62841_1
Attachment Originator..... :	DEKRA Certification B.V.
Master Attachment..... :	2021-08-27

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	CENELEC COMMON MODIFICATIONS (EN)		–
8	**MARKINGS AND INSTRUCTIONS**		–
8.4	Replace the 2nd paragraph with the following:		–
	Markings specified in 8.2 and 8.3 are clearly discernible from the outside of the tool.		P
	Other markings on the tool may be visible after removal of a cover, provided that the location of the markings is readily accessible.		P
8.14	The words "Original instructions" appear on the language version(s) verified by the manufacturer or his authorised representative.		P
	Where no "Original instructions" exist in the official language(s) of the country where the tool is to be used, a translation into that/those language(s) is provided by the manufacturer or his authorised representative or by the person bringing the tool into the language area in question.		P
	The translations bear the words "Translation of the original instructions", and they are accompanied by a copy of the "Original instructions".		P
8.14.2	The noise emission, which is measured in accordance with I.2		P
	Where LpA does not exceed 70 dB(A), this fact is indicated; L_{pA} [dB(A)] ..... : K_{pA} [dB(A)] ..... :	45dB	P
	A-weighted sound pressure level LpA and its uncertainty KpA, where LpA exceeds 70 dB(A).		N/A
	peak C-weighted instantaneous sound pressure value LpCpeak, where this exceeds 63 Pa (130 dB in relation to 20 µPa) L_{pCpeak} (dB) ..... : K_{pCpeak} (dB) ..... :		N/A

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Clause	Requirement + Test	Result - Remark	Verdict
	A-weighted sound power level LWA and its uncertainty KWA, where the A-weighted sound pressure level LpA exceeds 80 dB(A); L_{WA} [dB(A)] .....: K_{WA} [dB(A)] .....:		N/A
	The vibration total value and its uncertainty which is measured in accordance with I.3.		N/A
	When the vibration total value does not exceed 2,5 m/s², this is stated. Work mode - vibration emission value a (m/s²)….: Uncertainty K (m/s²)……...:		N/A
	When the vibration total value exceeds 2,5 m/s², its value is given in the instructions.		N/A
	Information that the declared vibration total value has been measured in accordance with a standard test method and may be used for comparing one tool with another		N/A
	Information that the declared vibration total value may also be used in a preliminary assessment of exposure.		N/A
	A warning that the vibration emission during actual use of the power tool can differ from the declared total value depending on the ways in which the tool is used		N/A
	A warning of the need to identify safety measures to protect the operator that are based on an estimation of exposure in the actual conditions of use (taking account of all parts of the operating cycle such as the times when the tool is switched off and when it is running idle in addition to the trigger time).		N/A

18	**ABNORMAL OPERATION**		N/A
18.8.1	In Table 4, replace the table footnote by the following: * Performance levels are to be specified in the relevant part of EN 62841-2, EN 62841-3 or EN 62841-4.		N/A
	Delete the 5th paragraph and the subsequent NOTE 3.		N/A

21	**CONSTRUCTION**		**–**
21.18.1	delete the 2nd paragraph.		N/A
21.18.1.Z1	Unless hand-held tools are equipped with a momentary power switch without lock-on device, voltage recovery following an interruption of the supply do not give rise to a hazard.		N/A

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Clause	Requirement + Test	Result - Remark	Verdict
	The relevant part of EN 62841-2 specifies if this subclause applies and gives specific requirements.		N/A
21.18.2.1	Unless transportable tools are equipped with a momentary power switch without lock-on device, voltage recovery following an interruption of the supply do not give rise to a hazard.		N/A
	The relevant part of EN 62841-3 specifies if this subclause applies and gives specific requirements.		N/A
ANNEX E	**METHODS OF APPLYING ISO 13849-1 TO POWER TOOLS**		**–**
	(Void)		N/A
ANNEX I	**MEASUREMENT OF NOISE AND VIBRATION EMISSIONS**		**–**
	Replace the title of Annex I by the following **ANNEX I – (NORMATIVE)**		N/A
I.2	Noise test code (grade 2)		N/A
I.2.Z1	Noise reduction		–
	Add the following before I.2.1:		
	Noise reduction at tools is an integral part of the design process and is achieved by particularly applying measures at source to control noise, see for example EN ISO 11688-1.		N/A
	The success of the applied noise reduction measures is assessed on the basis of the actual noise emission values in relation to other machines of the same type with comparable non acoustical technical data.		N/A
	The major sound sources of tools are: motor, fan, gear.		N/A
I.2.1	General		–
	The noise emission determined by using a machine which has design and technical specifications replicating the machine concerned.		N/A
	The overall noise can be divided into the pure machine noise and the noise generated from the processed workpiece.		N/A
	The load conditions for particular tools are therefore specified in the relevant part of IEC 62841-2, IEC 62841-3 or IEC 62841-4.		N/A
I.2.2	Sound power level measured according to ISO 3744		N/A
I.2.2.2	Hand-held power tools		N/A
	For all hand-held power tools, the sound power level is determined by using a hemispherical / cylindrical measurement surface according to Figure I.2.		N/A

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Clause	Requirement + Test	Result - Remark	Verdict

I.2.2.3	For all transportable power tools, the sound power level is determined by using a cubic measurement surface according to Figure I.3.		N/A
I.2.2.4	The sound power level of lawn and garden machinery is determined as specified in the relevant part of IEC 62841-4.		N/A
I.2.3	Emission sound pressure level determination		–
I.2.3.1	The A-weighted emission sound pressure level of hand-held tools at the work station LpA according to ISO 11203		N/A
	If required, LpCpeak is measured at each of the five measurement positions specified in I.2.2		N/A
I.2.3.2	The A-weighted emission sound pressure level of transportable tools at the work station, LpA, is determined according to ISO 11201, grade 2.		N/A
	If required, the C-weighted peak emission sound pressure level LpCpeak is measured at the same operator's position as the A-weighted sound pressure level LpA.		N/A
I.2.3.3	The emission sound pressure level of lawn and garden machinery is determined as specified in the relevant part of IEC 62841-4.		N/A
I.2.4	Installation and mounting conditions of the power tools during noise tests		–
	The power tool under test is new and equipped with accessories which affect the acoustic properties, as recommended by the manufacturer.		N/A
	Prior to commencing testing, the power tool (including any required ancillary equipment) is set up in a stable condition in accordance with the manufacturer's instructions for safe use.		N/A
	A hand-held tool is held by the operator or suspended in such a way as to correspond to normal use, as specified in the relevant part of IEC 62841-2		N/A
	A transportable tool is so positioned, either placed on the test bench of Figure I.1		N/A
	Lawn and garden machinery is used and positioned as specified in the relevant part of IEC 62841-4.		N/A
I.2.5	Operating conditions		–
	Tools are tested under the two operating conditions "no-load" or "load" as appropriate for the type of tool and specified in the relevant part of IEC 62841-2, IEC 62841-3 or IEC 62841-4.		N/A
I.2.6	Measurement uncertainties		–
	Uncertainties according to standard determined, recorded and reported		N/A

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Clause	Requirement + Test	Result - Remark	Verdict

I.2.7	Information and deviations are recorded.		N/A
I.2.8	Information to be reported		N/A
	- reference to this noise test code / basic standard		N/A
	- description of the power tool;		N/A
	- description of mounting and operating conditions		N/A
	Sound power level L_{WA} (dB(A)).....:		N/A
	Sound pressure level L_{PA} (dB(A)).....:		N/A
	C-weighted peak emission sound pressure level LpCpeak (dB).....:		N/A
I.2.9	Declaration and verification of noise emission values		N/A
	Sound power level LWA (dB(A)).....:		N/A
	Sound pressure level LPA (dB(A)).....:		N/A
	C-weighted peak emission sound pressure level LpCpeak (dB).....:		N/A
I.3	Vibration		–
I.3.Z1	Vibration reduction		N/A
	Add the following before I.3.1:		N/A
	The vibration at the handles is kept as low as possible without unduly affecting the performance and the ergonomics (weight, handling, etc.) of the tool.		N/A
	In particular vibration is reduced by the application of engineering measures as given in CR 1030-1.		N/A
	The success of the applied vibration measures is assessed by comparing the vibration levels for the tool with those for other tools of the same type and with a comparable specification and performance.		N/A
I.3.1	Vibration measurement – General		N/A
	Details for particular types of tools are given in the relevant part of IEC 62841-2, IEC 62841-3 or IEC 62841-4.		N/A
	The vibration total value may be determined by using the measurements from a machine which has design and technical specifications replicating the machine concerned.		N/A
I.3.2	Symbols		N/A
I.3.3	Characterization of vibration		N/A
I.3.3.1	Direction of measurement		N/A
	Directions may be defined in the relevant part of IEC 62841-2, IEC 62841-3 or IEC 62841-4.		N/A
	If not defined the three orthogonal directions X, Y and Z as shown in Figure I.4. are related		N/A
I.3.3.2	Location of measurement		N/A

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Clause	Requirement + Test	Result - Remark	Verdict
	The measurement positions for particular types of tools are specified in the relevant part of IEC 62841-2, IEC 62841-3 or IEC 62841-4.		N/A
I.3.3.3	Magnitude of vibration		N/A
I.3.3.4	Combination of vibration directions		N/A
I.3.4	Instrumentation requirements		N/A
	The vibration measurement equipment is in accordance with ISO 8041.		N/A
	Instrumentation for measuring other parameters whose characteristics are not covered by ISO 8041, is specified in the relevant part of IEC 62841-2, IEC 62841-3 or IEC 62841-4.		N/A
I.3.4.2	Transducers		N/A
I.3.5	Testing and operating conditions of the tool		N/A
I.3.5.1	Replace the 4th paragraph with the following:		N/A
	When the test procedure is not provided in a relevant part of EN 62841-2, EN 62841-3 or EN 62841-4, an operating condition is specified that is reproducible and representative of the noisiest operation in typical usage of the machine.		N/A
	The vibration test may simulate a single phase of a task or a working cycle, consisting of a set of operations where the operator is being exposed to vibration.		N/A
	However, the operating condition for the noise emission test is, if practicable, also used for the vibration test.		N/A
I.3.5.2	Attachment, workpiece and task		N/A
	Details for task and workpiece are given in the relevant part of IEC 62841-2, IEC 62841-3 or IEC 62841-4.		N/A
I.3.5.3	Operating conditions		N/A
	The relevant part of IEC 62841-2, IEC 62841-3 or IEC 62841-4 describes the modes of operation and the calculation of the declared emission value.		N/A
I.3.5.4	Operator		N/A
I.3.6	Measurement procedure and validity		N/A
I.3.6.1	Reported vibration values		N/A
	Details are specified in the relevant part of IEC 62841-2, IEC 62841-3 or IEC 62841-4.		N/A
	Work mode - vibration emission value a (m/s²)….:		N/A
	Uncertainty K (m/s²)……..:		N/A
I.3.6.2	Declaration of the vibration total value		N/A

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Clause	Requirement + Test	Result - Remark	Verdict
	If required by the relevant part of IEC 62841-2, IEC 62841-3 or IEC 62841-4, the work mode description corresponding to the vibration emission is stated next to each declared value.		N/A
	Work mode - vibration emission value a (m/s²)….:		N/A
	Uncertainty K (m/s²)……..…...:		N/A
I.3.7	Measurement report		N/A
	The report includes the following information:		N/A
	a) reference to this standard		N/A
	b) specification of the machine		N/A
	c) attachments or accessories;		N/A
	d) operating and testing conditions		N/A
	e) measuring institution		N/A
	f) date of measurement		N/A
	g) instrumentation		N/A
	h) position and fastening of transducers, measuring directions and individual vibration values when relevant		N/A
	i) the arithmetic mean total vibration ah, for each operator the total vibration value ahv and the three single axes weighted acceleration values ahw. It is good practice to report all the measured values		N/A
	j) the uncertainty K of the vibration total value ah.		N/A
	Any deviations from the vibration test code in this standard is reported together with the technical justification for such deviations.		N/A

ANNEX K	**BATTERY TOOLS AND BATTERY PACKS**		**N/A**
K8.14.2 Z1	For battery tools with integral battery: instruction, how the integral battery can be removed safely from the tool after the tool's end of life, and information about the type of battery such as Li-Ion, NiCd and NiMH.		N/A

ANNEX L	**BATTERY TOOLS AND BATTERY PACKS PROVIDED WITH MAINS CONNECTION OR NON-ISOLATED SOURCES**		**N/A**
K8.14.2 Z1	For battery tools with integral battery: instruction, how the integral battery can be removed safely from the tool after the tool's end of life, and information about the type of battery such as Li-Ion, NiCd and NiMH.		N/A

Pictures

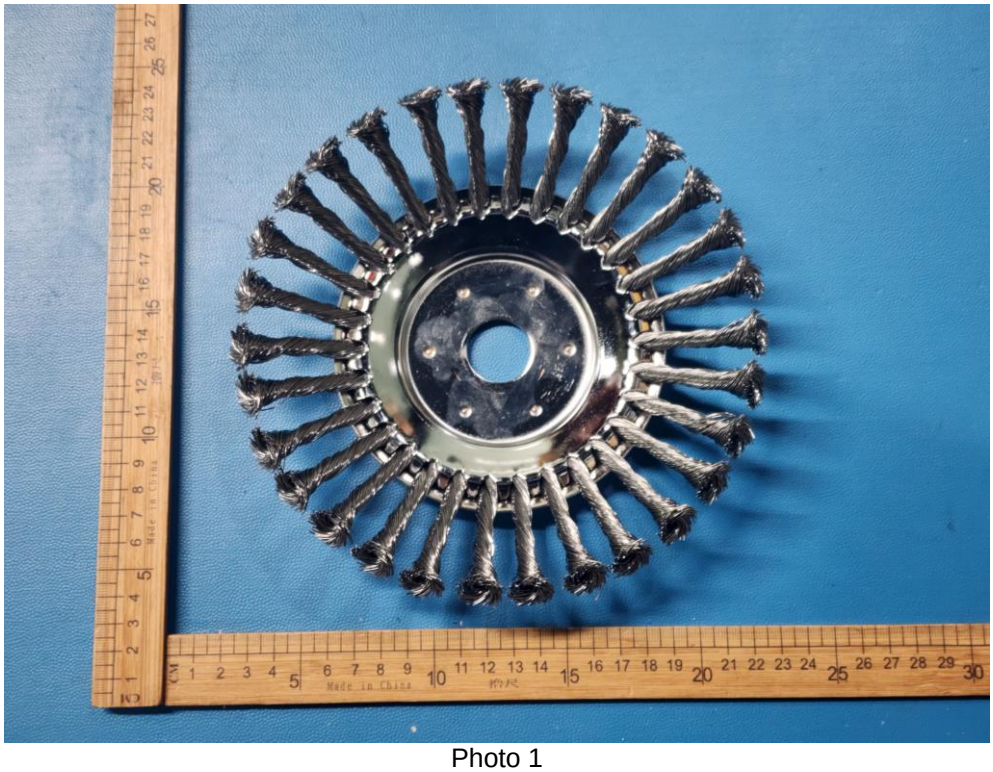


Photo 1

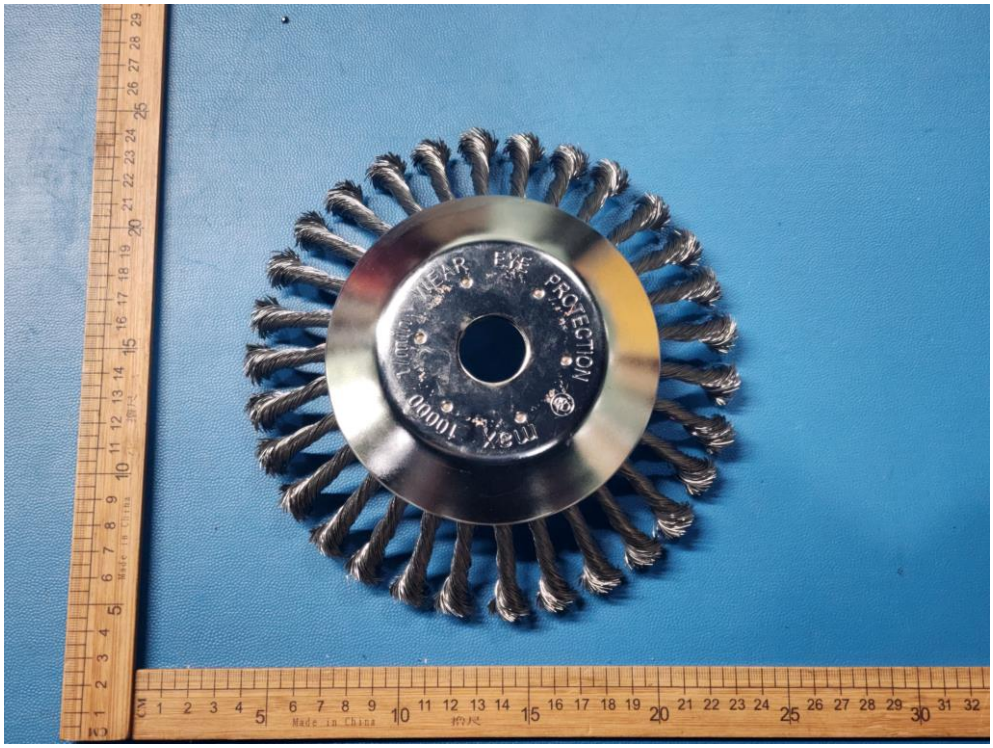


Photo 2

End of Report